

Understanding Large Language Models

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Session 05: Finetuning & RLHF

Main learning goals

1. Recap

- model sizes, capabilities

2. Supervised Finetuning

- basic concepts, PEFT, LoRA, QLoRA, prompt tuning

3. Instruction Finetuning

- basic concepts

4. Recap: RL

- reinforcement learning

5. Learning from Humans...

- online vs. offline learning, quality signals

6. RLHF

- language models as policies, cutting the human out of the loop



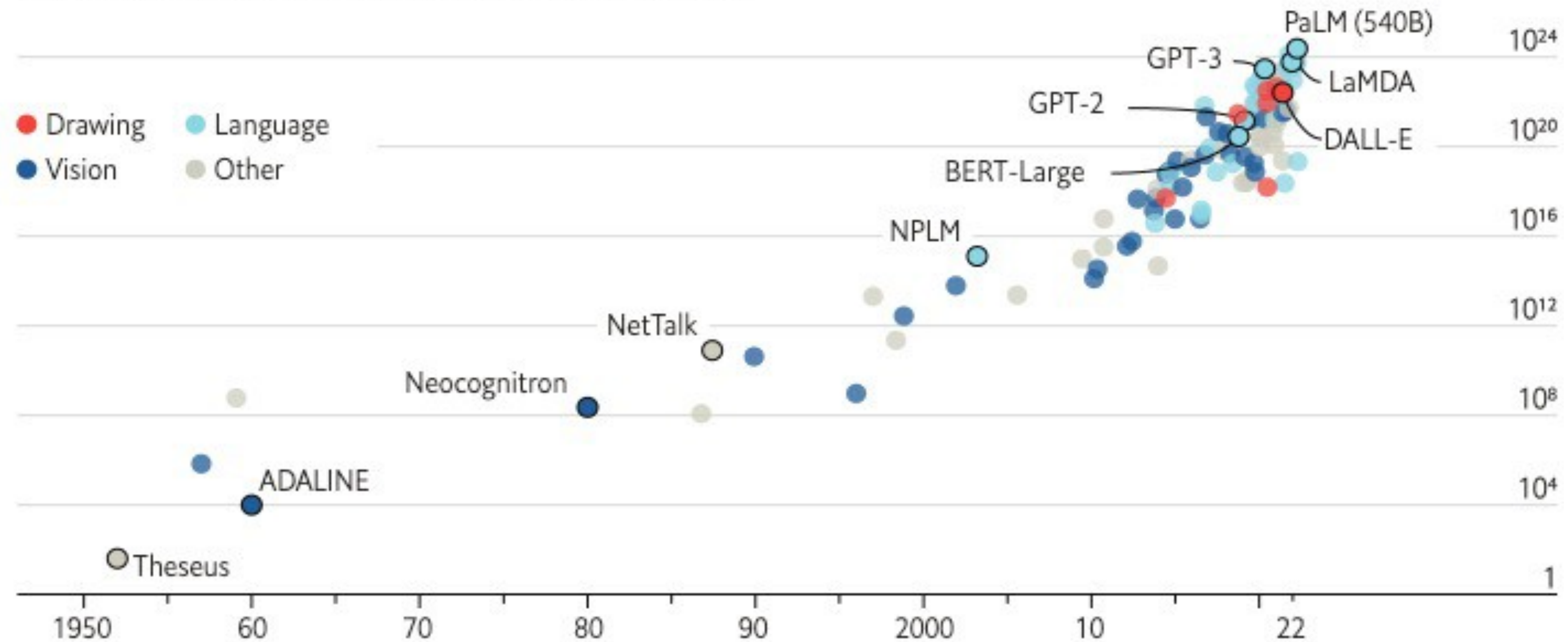
Recap

Larger and larger models

The blessings of scale

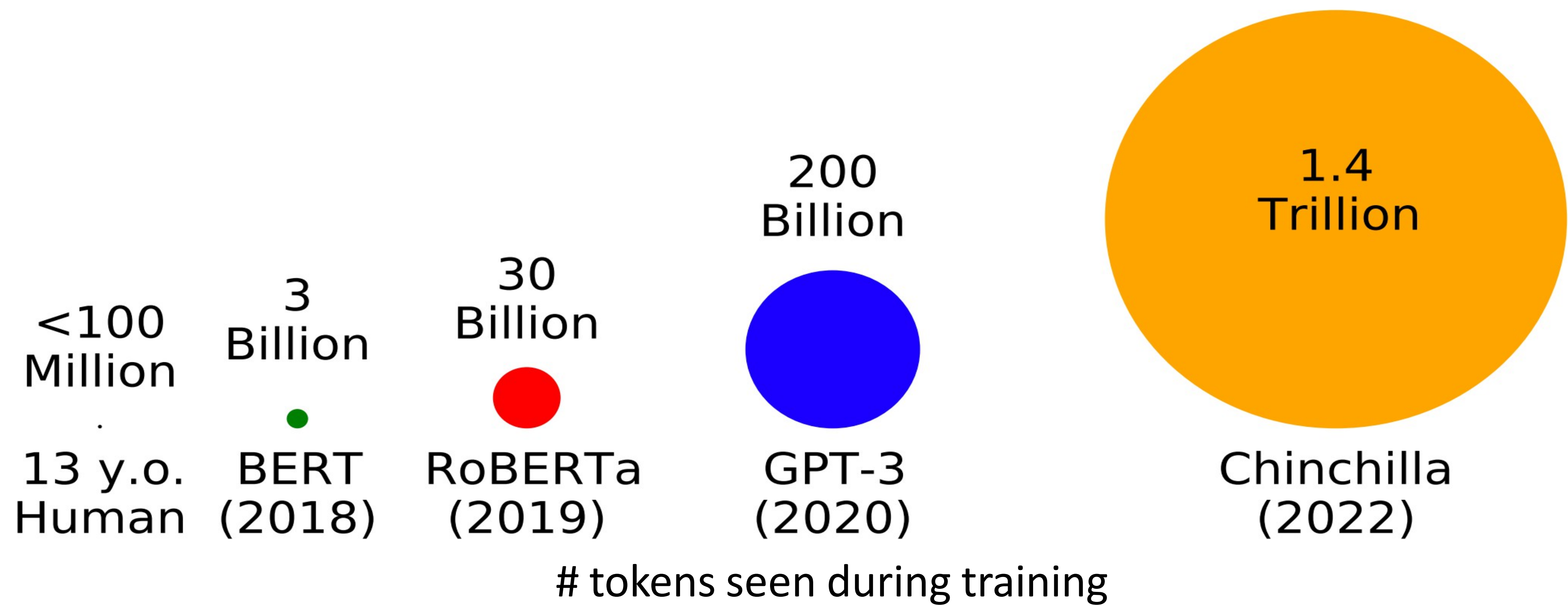
AI training runs, estimated computing resources used

Floating-point operations, selected systems, by type, log scale



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

Trained on More and More Data



<https://babylm.github.io/>

What does Pretraining Teach us?

- *Stanford University is located in _____, California. [Trivia]*
- *I put ____ fork down on the table. [syntax]*
- *The woman walked across the street, checking for traffic over ____ shoulder. [coreference]*
- *I went to the ocean to see the fish, turtles, seals, and _____. [lexical semantics/topic]*
- *Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was _____. [sentiment]*
- *Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the _____. [some reasoning – this is harder]*
- *I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, _____ [some basic arithmetic; they don't learn the Fibonacci sequence]*

Rudimentary Models of Agents, Beliefs, and Actions

Pat watches a demonstration of a bowling ball and a leaf being dropped at the same time in a vacuum chamber. Pat, who is a physicist, predicts that the bowling ball and the leaf will fall at the same rate.

Changing the last sentence of the prompt, we get:

...Pat, who has never seen this demonstration before, predicts that the bowling ball will fall to the ground first. This is incorrect. In a vacuum chamber, there is no air

Language Models as Agent Models [[Andreas, 2022](#)]

Learning to Code (Pretty well!)

```
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8     const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9         method: "POST",
10        body: `text=${text}`,
11        headers: {
12            "Content-Type": "application/x-www-form-urlencoded",
13        },
14    });
15    const json = await response.json();
16    return json.label === "pos";
17 }
```


Memorizing Advanced Medical Facts

Rapid and chronic ethanol tolerance are composed of distinct memory-like states in Drosophila

Abstract

Ethanol tolerance is the first type of behavioral plasticity and neural plasticity that is induced by ethanol intake, and yet its molecular and circuit bases remain largely unexplored. Here, we characterize three distinct forms of ethanol tolerance in male Drosophila: rapid, chronic, and repeated. Rapid tolerance is composed of two short-lived memory-like states, one that is labile and one that is consolidated. Chronic tolerance, induced by continuous exposure, lasts for two days, induces ethanol preference, and hinders the development of rapid tolerance through the activity of

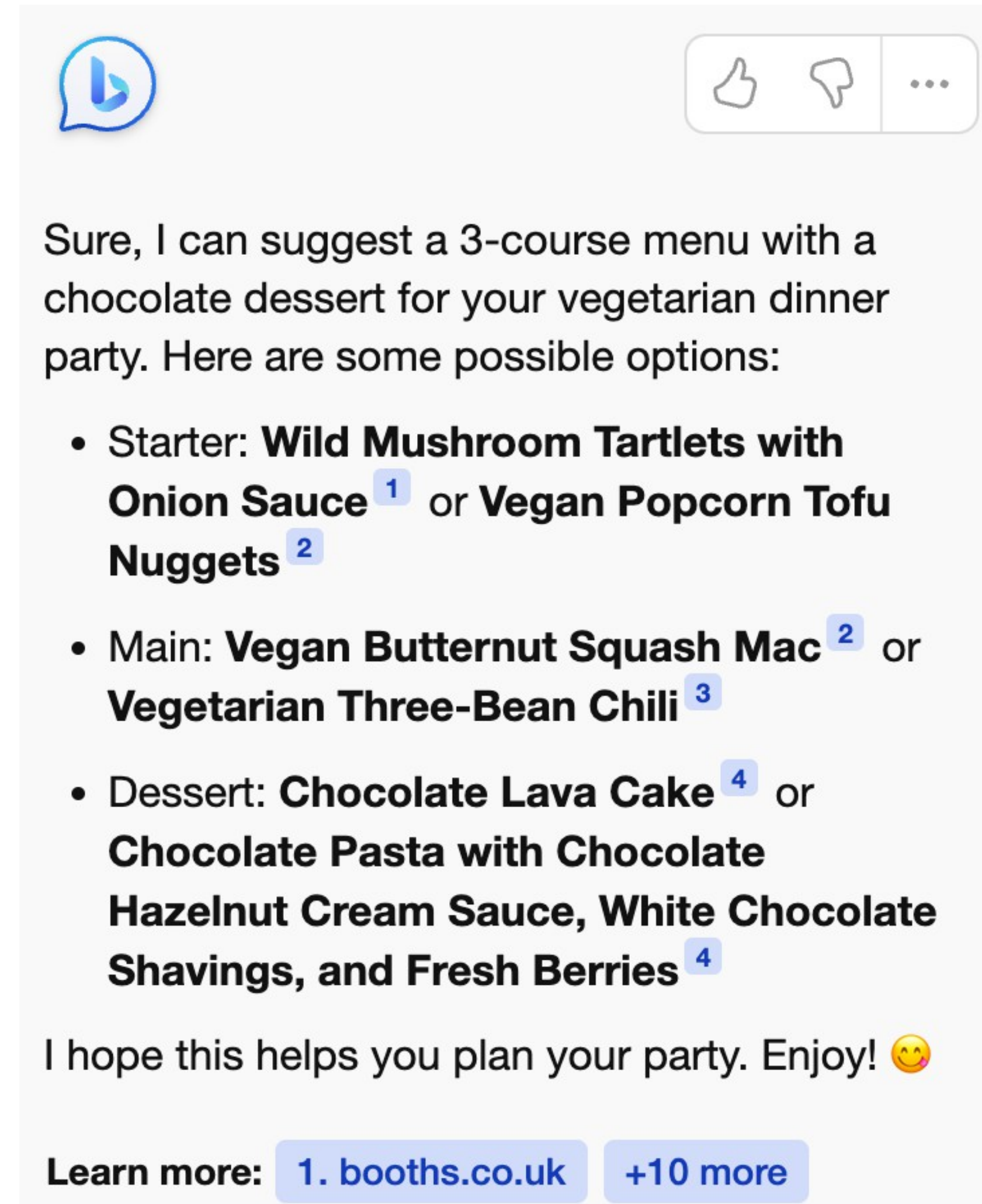
[[Larnerd, 2023](#)]

Multi-Step Problem Solving

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

[[Microsoft Bing](#)]

(Also see OpenAI's ChatGPT, Google's Bard, Anthropic's Claude)



The image shows a screenshot of a Microsoft Bing chat interface. At the top left is the Bing logo, and at the top right are icons for thumbs up, thumbs down, and a menu. The chat text reads: "Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:" followed by a bulleted list of three options. Each option has a small blue square with a number (1, 2, or 4) next to it. The first option is "Starter: Wild Mushroom Tartlets with Onion Sauce¹ or Vegan Popcorn Tofu Nuggets²". The second option is "Main: Vegan Butternut Squash Mac² or Vegetarian Three-Bean Chili³". The third option is "Dessert: Chocolate Lava Cake⁴ or Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries⁴". Below the list, it says "I hope this helps you plan your party. Enjoy! 😊". At the bottom, there is a "Learn more:" label followed by two buttons: "1. booths.co.uk" and "+10 more".

Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:

- Starter: **Wild Mushroom Tartlets with Onion Sauce¹** or **Vegan Popcorn Tofu Nuggets²**
- Main: **Vegan Butternut Squash Mac²** or **Vegetarian Three-Bean Chili³**
- Dessert: **Chocolate Lava Cake⁴** or **Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries⁴**

I hope this helps you plan your party. Enjoy! 😊

Learn more: [1. booths.co.uk](#) [+10 more](#)

Multitask Assistants

- How do we get from *this*
 - *Stanford University is located in _____*
- to *this?*

ChatGPT		
 Examples	 Capabilities	 Limitations
"Explain quantum computing in simple terms"	Remembers what user said earlier in the conversation	May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?"	Allows user to provide follow-up corrections	May occasionally produce harmful instructions or biased content
"How do I make an HTTP request in Javascript?"	Trained to decline inappropriate requests	Limited knowledge of world and events after 2021

Language Modeling $\neq$ Assisting Users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].

Language Modeling $\neq$ Assisting Users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION **Human**

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].

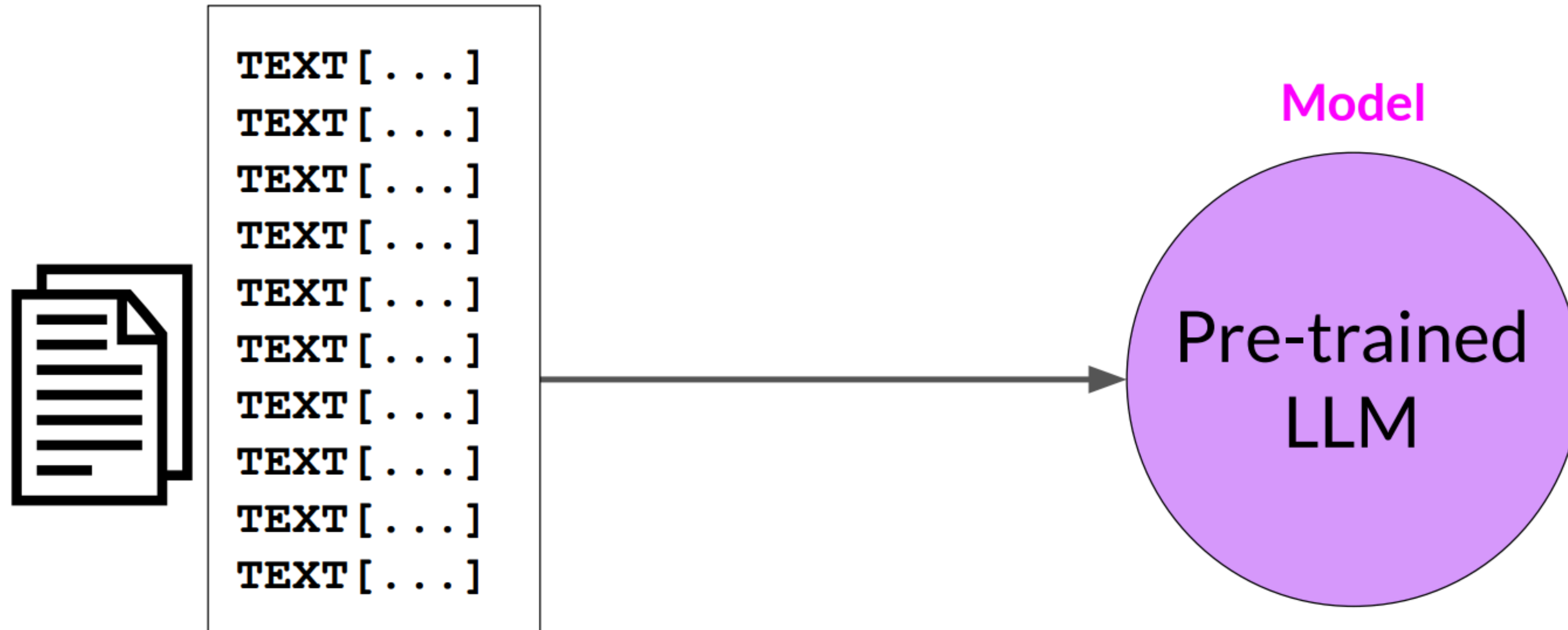
Finetuning to the rescue!



Supervised Finetuning

Pretraining

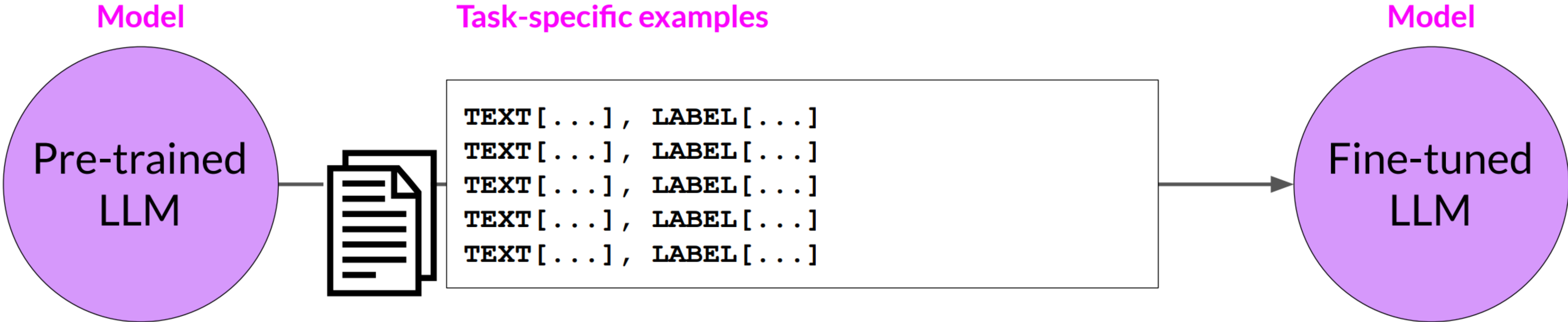
LLM pre-training



GB - TB - PB
of unstructured textual data

Conventional Finetuning

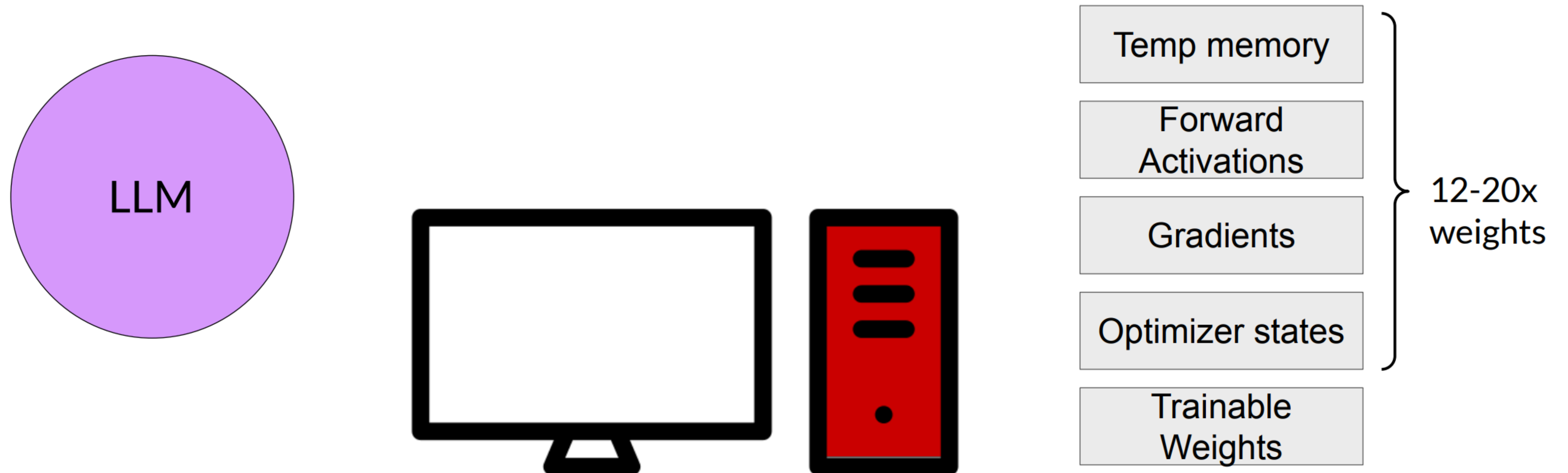
LLM fine-tuning



GB - TB
of labeled examples for a specific
task or set of tasks

Finetuning at the Frontier

Full fine-tuning of large LLMs is challenging



Getting Smart with Finetuning

PEFT methods summary

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

Additive

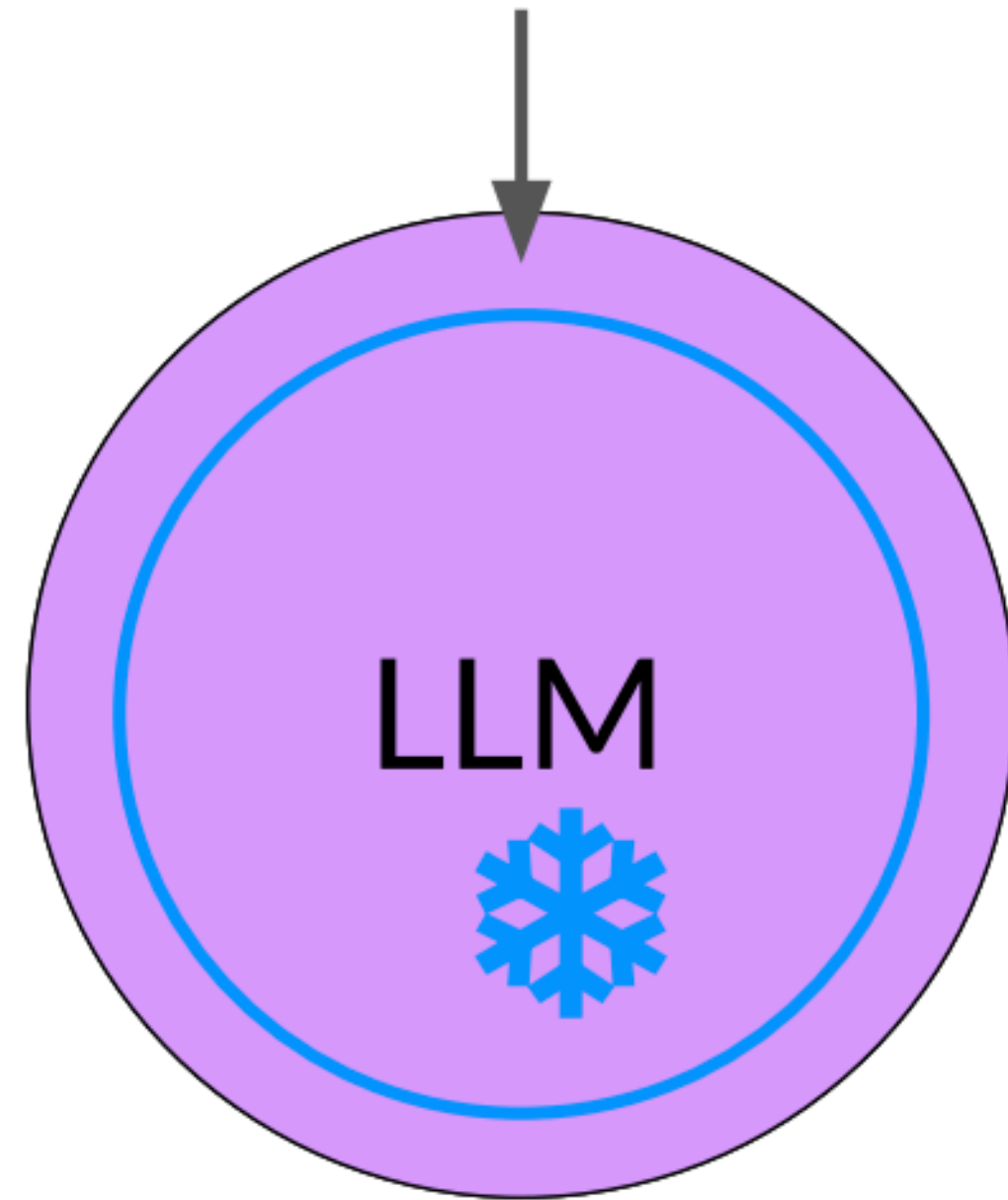
Add trainable layers or parameters to model

Adapters

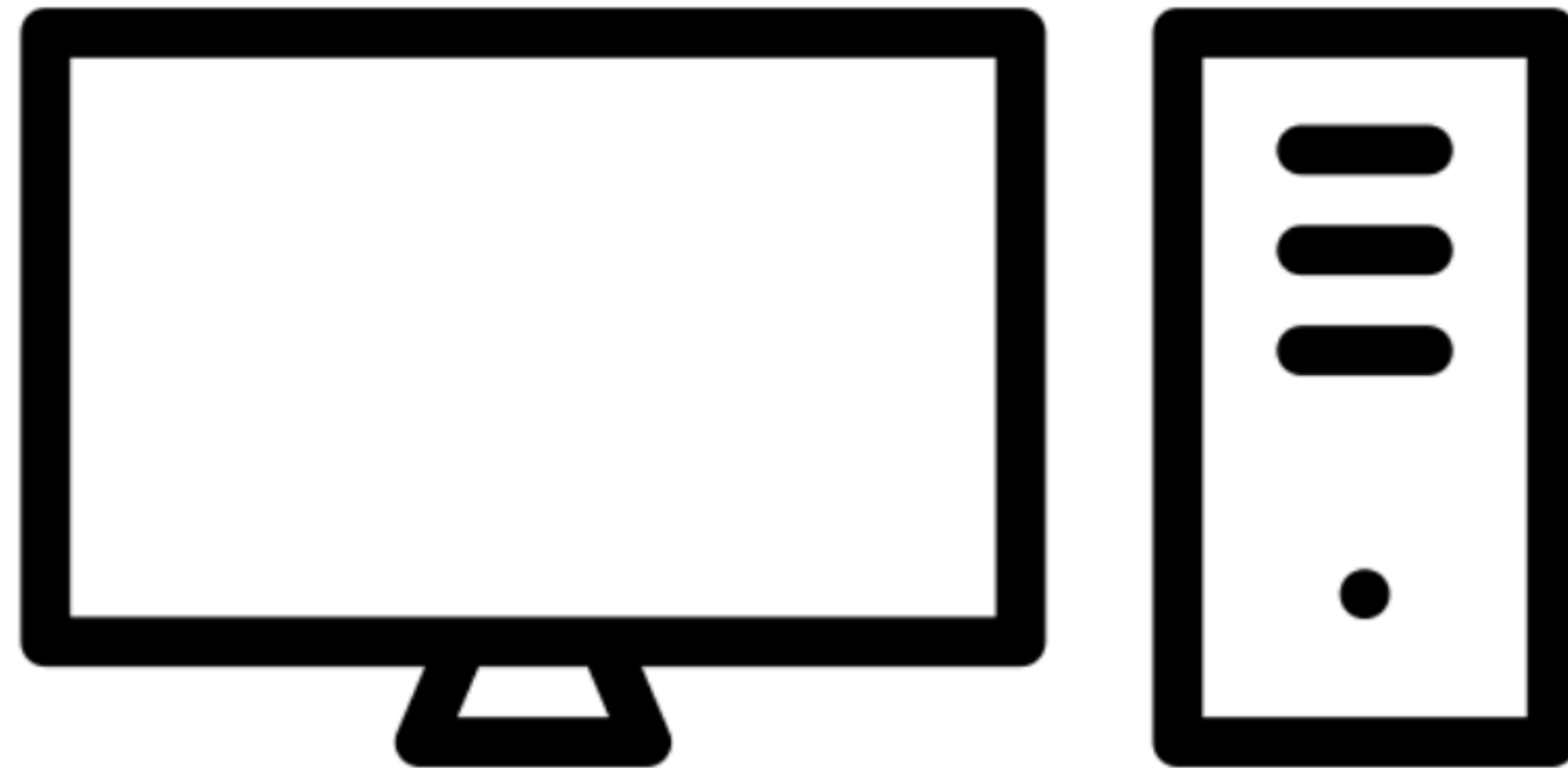
Soft Prompts
Prompt Tuning

Selective Finetuning

Small number of
trainable layers

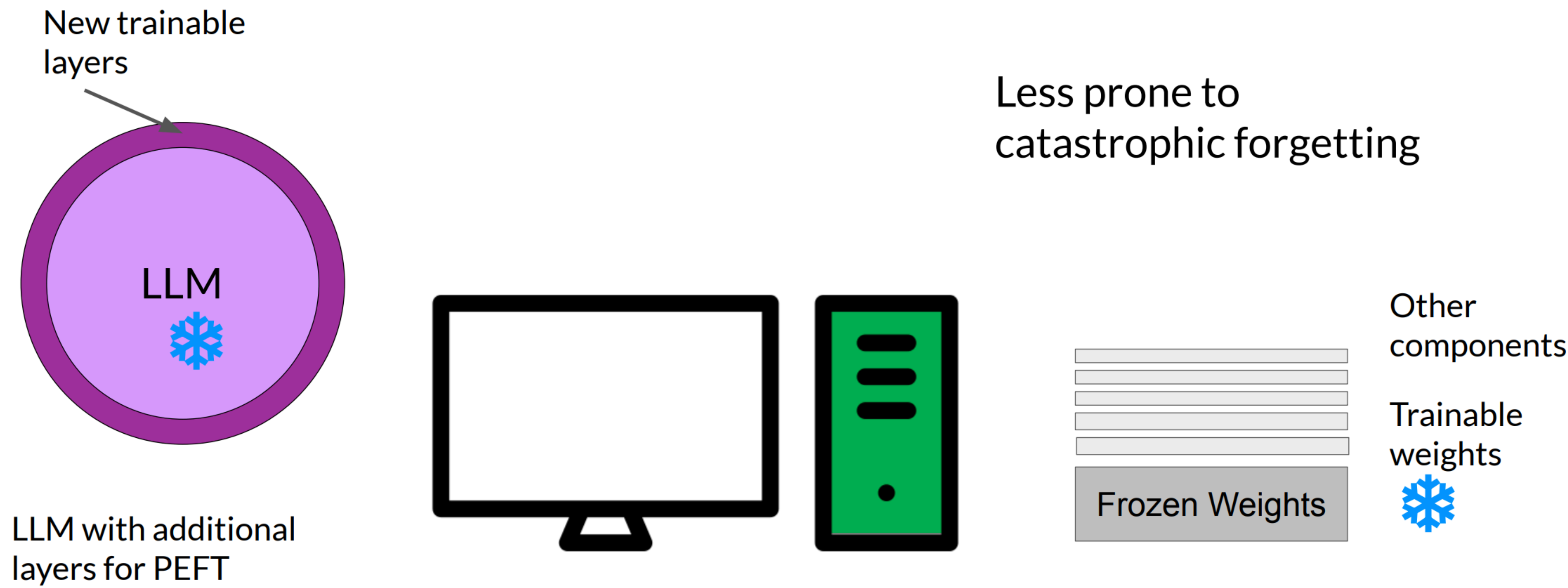


LLM with most layers
frozen



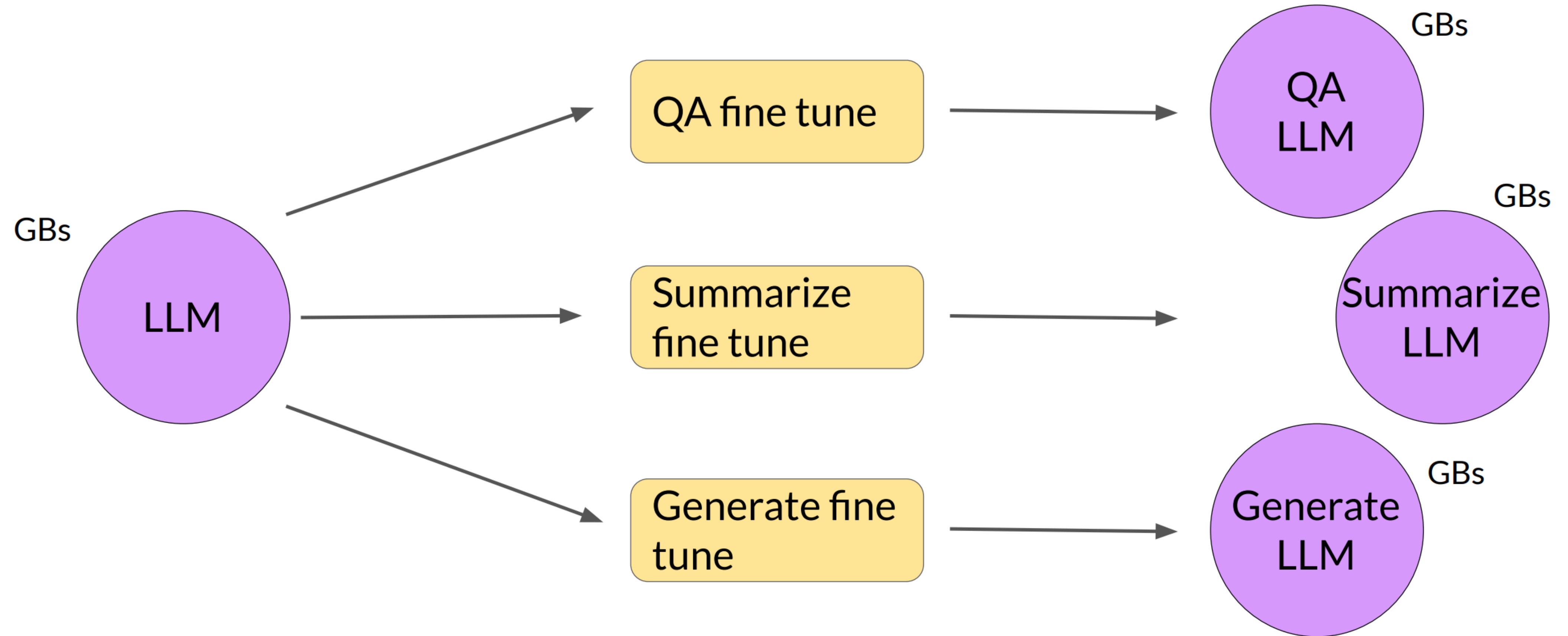
Selective Finetuning

Parameter efficient fine-tuning (PEFT)



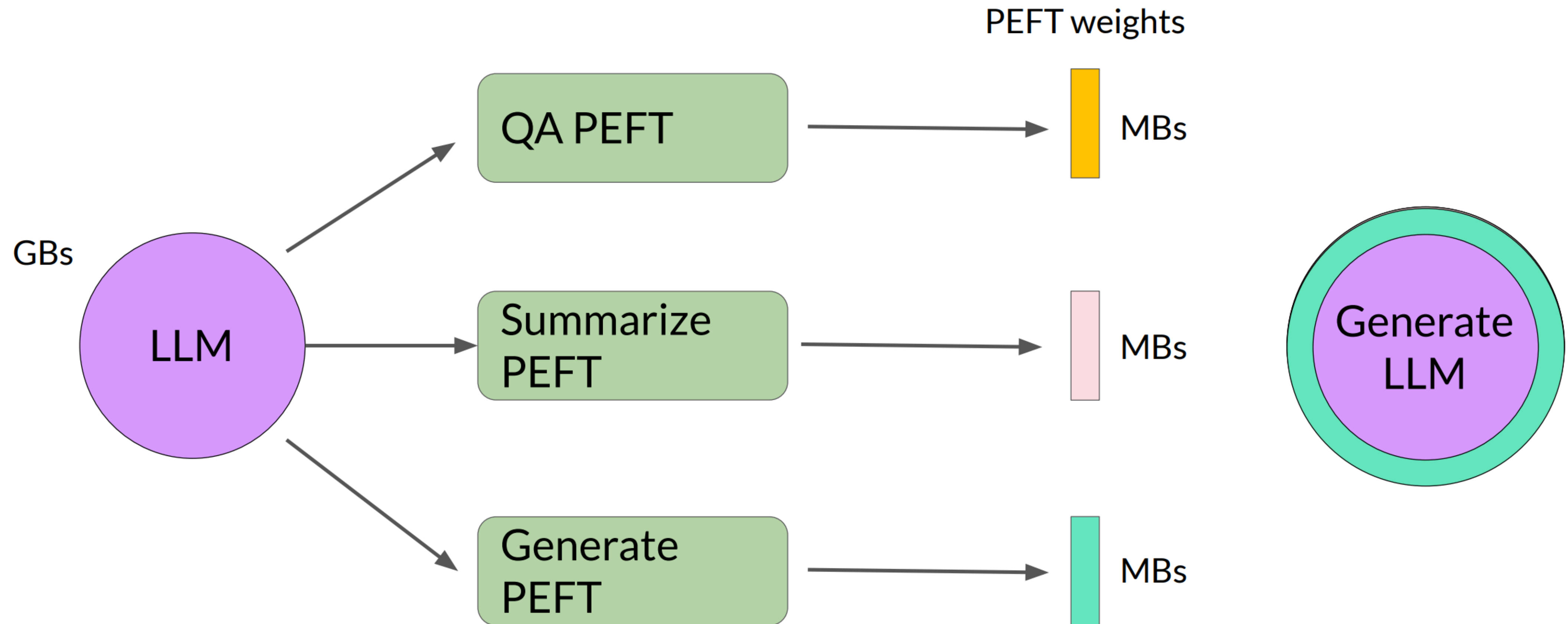
Selective Finetuning

Full fine-tuning creates full copy of original LLM per task



Selective Finetuning

PEFT fine-tuning saves space and is flexible



Getting Smart with Finetuning

PEFT methods summary

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

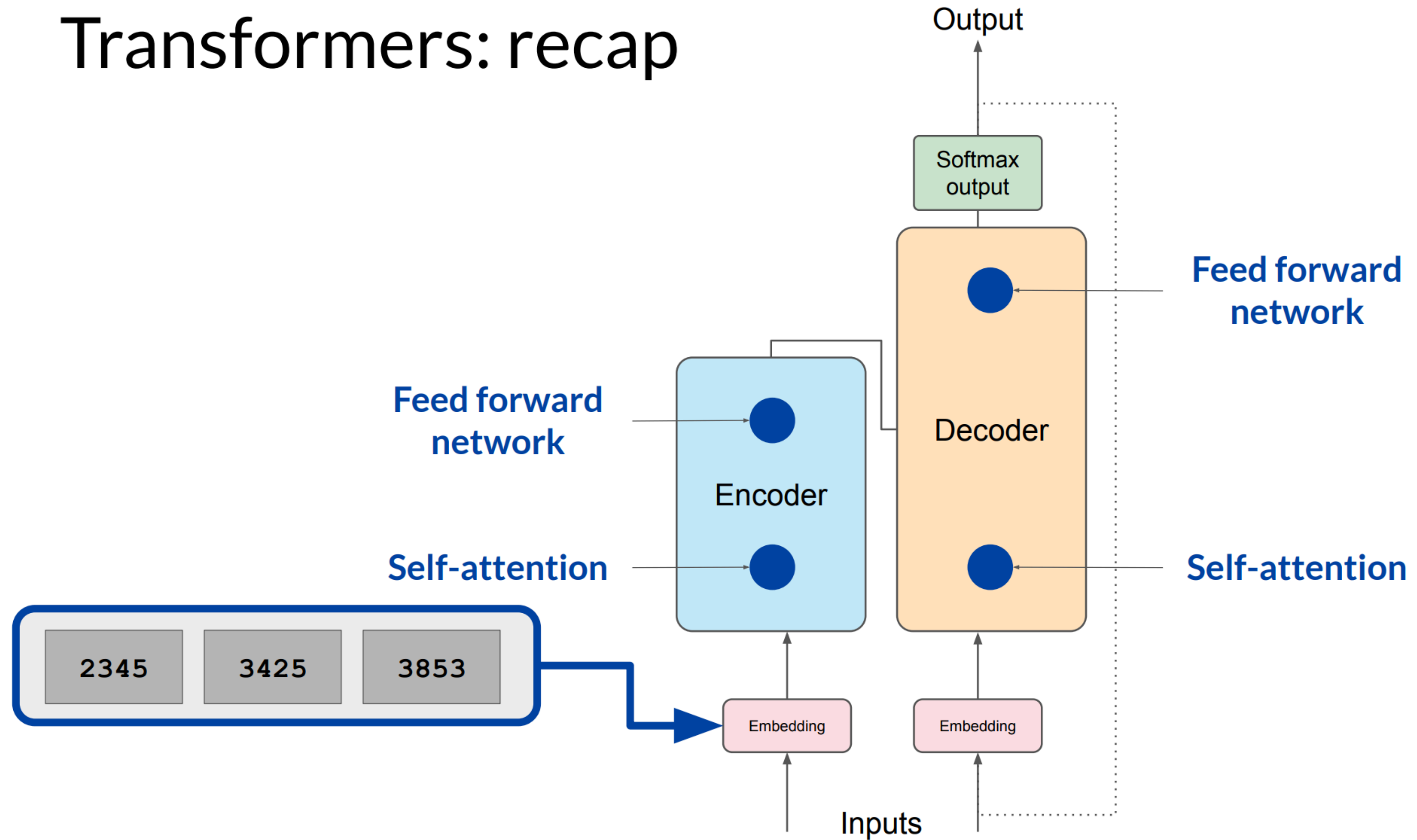
Additive

Add trainable layers or parameters to model

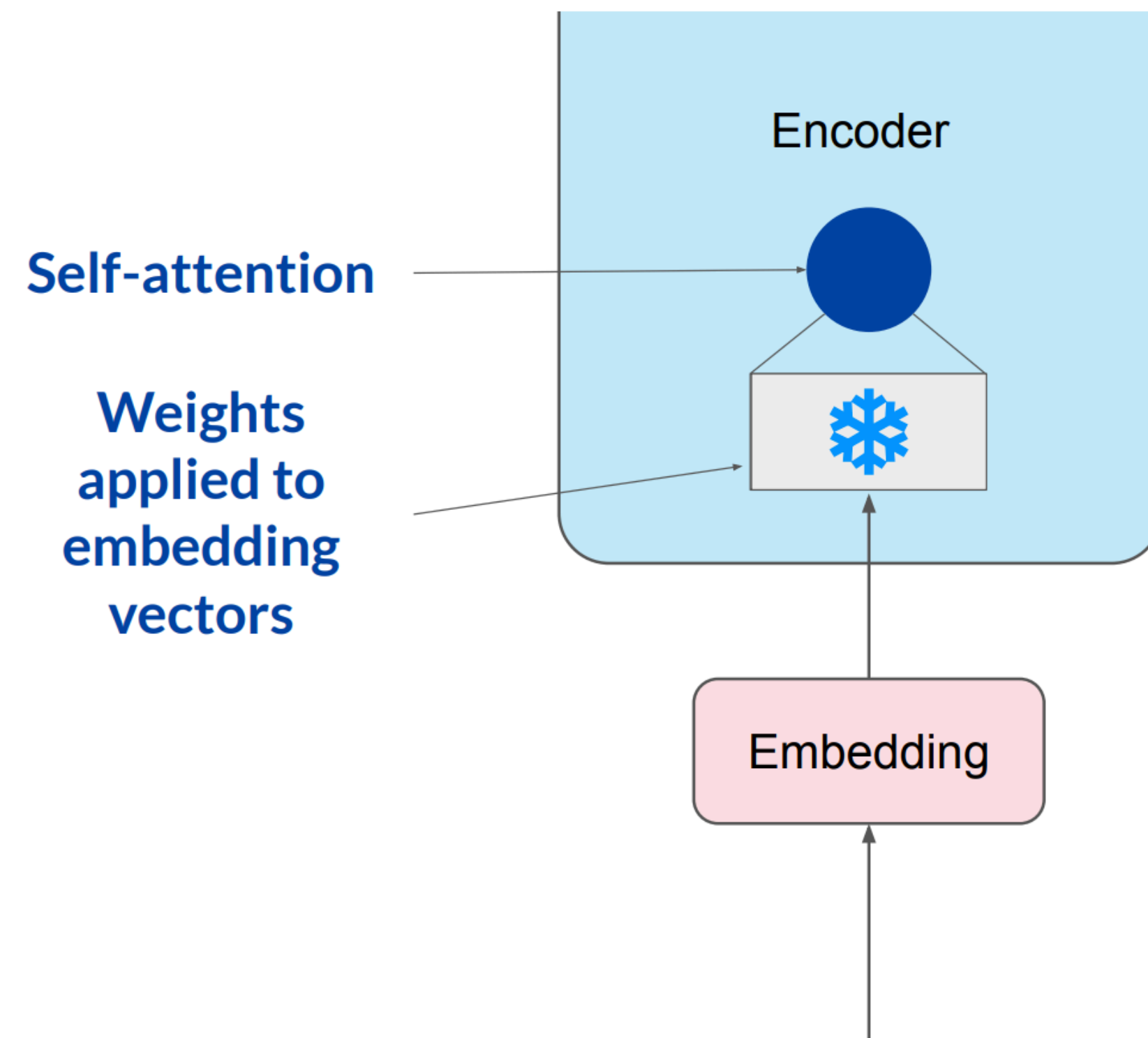
Adapters

Soft Prompts
Prompt Tuning

Transformers: recap

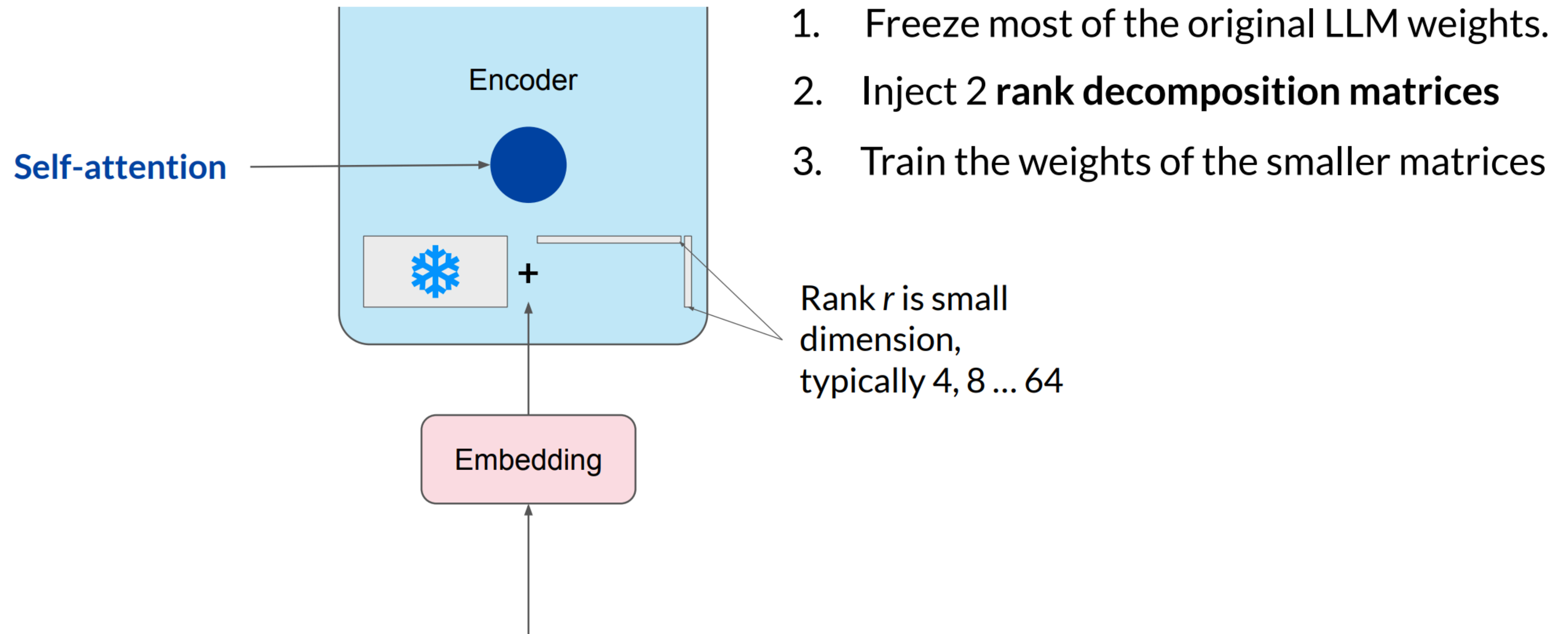


LoRA: Low Rank Adaption of LLMs

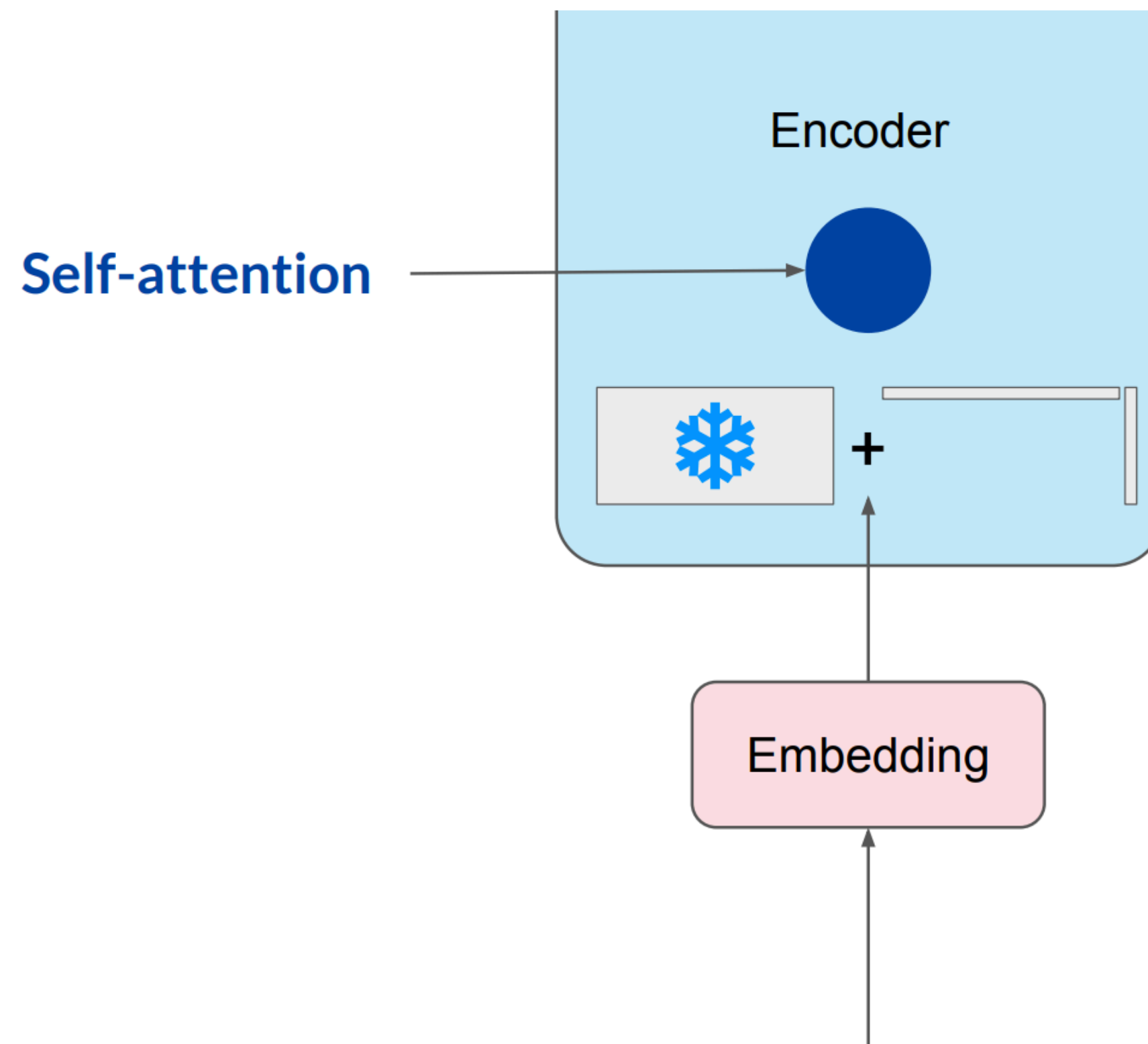


1. Freeze most of the original LLM weights.

LoRA: Low Rank Adaption of LLMs



LoRA: Low Rank Adaption of LLMs



1. Freeze most of the original LLM weights.
2. Inject 2 **rank decomposition matrices**
3. Train the weights of the smaller matrices

Steps to update model for inference

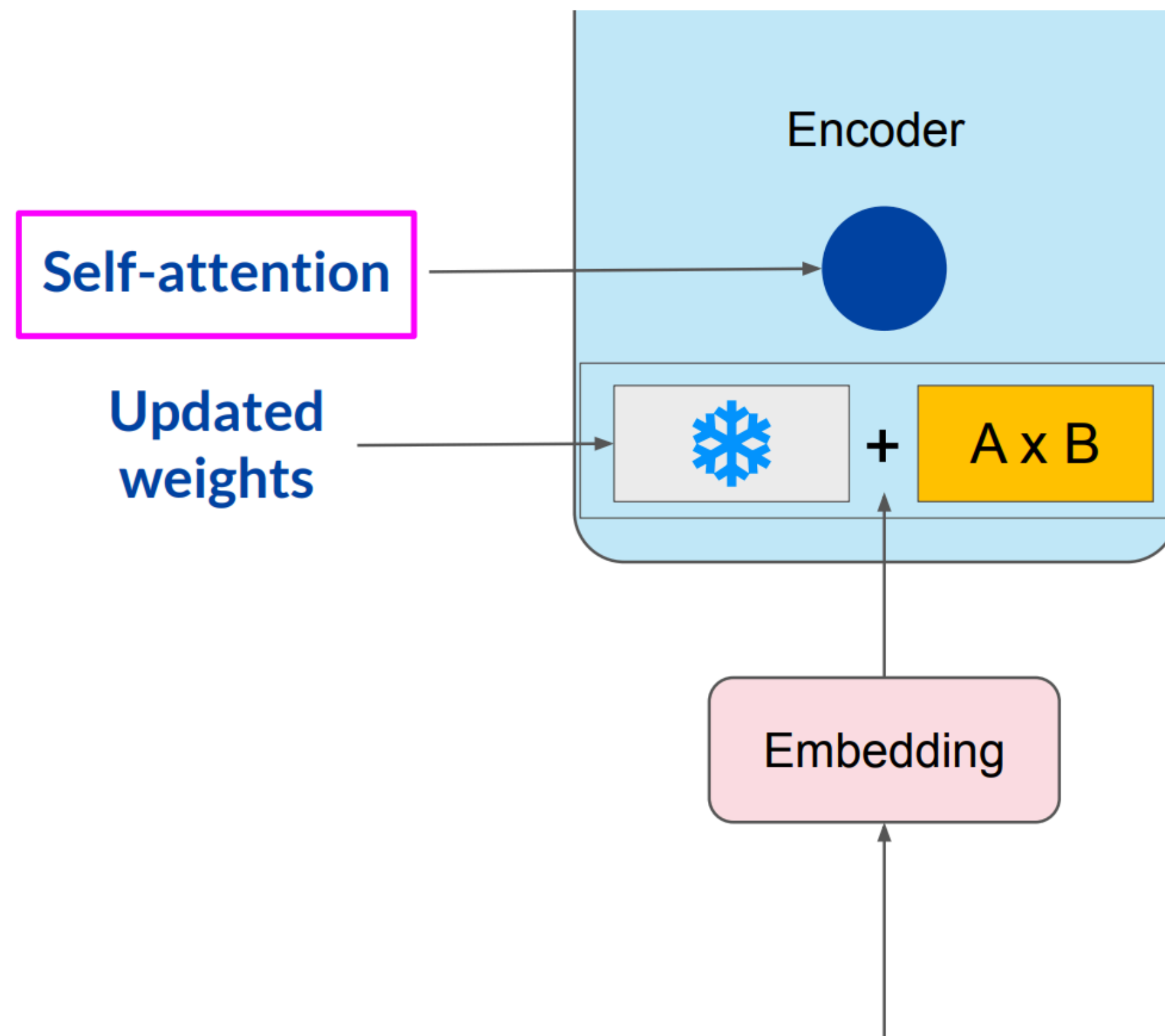
1. Matrix multiply the low rank matrices

$$\overline{B} * \begin{matrix} | \\ A \end{matrix} = \boxed{A \times B}$$

2. Add to original weights

$$\boxed{\begin{matrix} \text{snowflake} \end{matrix}} + \boxed{A \times B}$$

LoRA: Low Rank Adaption of LLMs



1. Freeze most of the original LLM weights.
2. Inject 2 **rank decomposition matrices**
3. Train the weights of the smaller matrices

Steps to update model for inference:

1. Matrix multiply the low rank matrices

$$\overline{B} * \begin{matrix} | \\ A \end{matrix} = \text{A x B}$$

2. Add to original weights

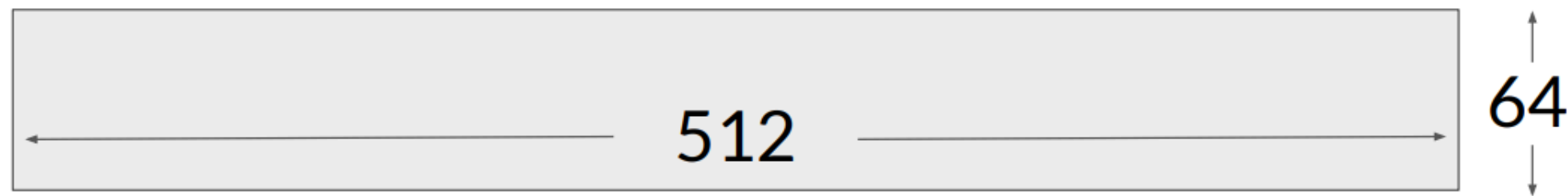
$$\begin{matrix} \text{Snowflake} \end{matrix} + \text{A x B}$$

LoRA

Concrete example using base Transformer as reference

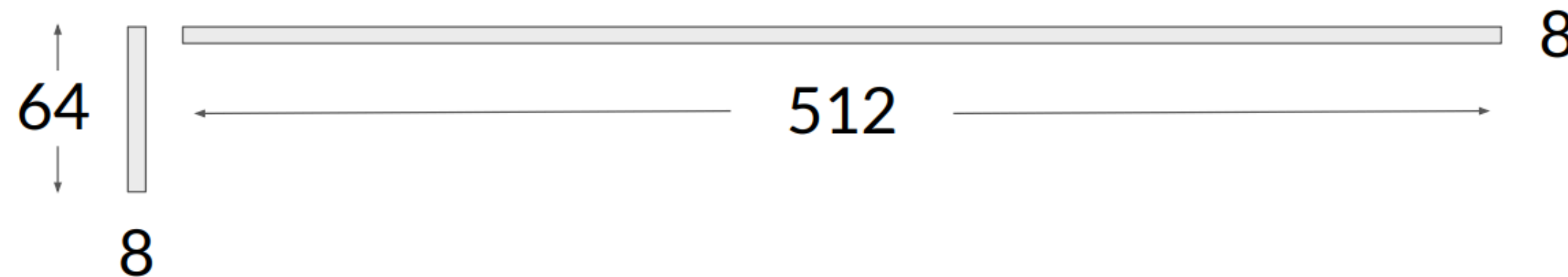
Use the base Transformer model presented by Vaswani et al. 2017:

- Transformer weights have dimensions $d \times k = 512 \times 64$
- So $512 \times 64 = 32,768$ trainable parameters



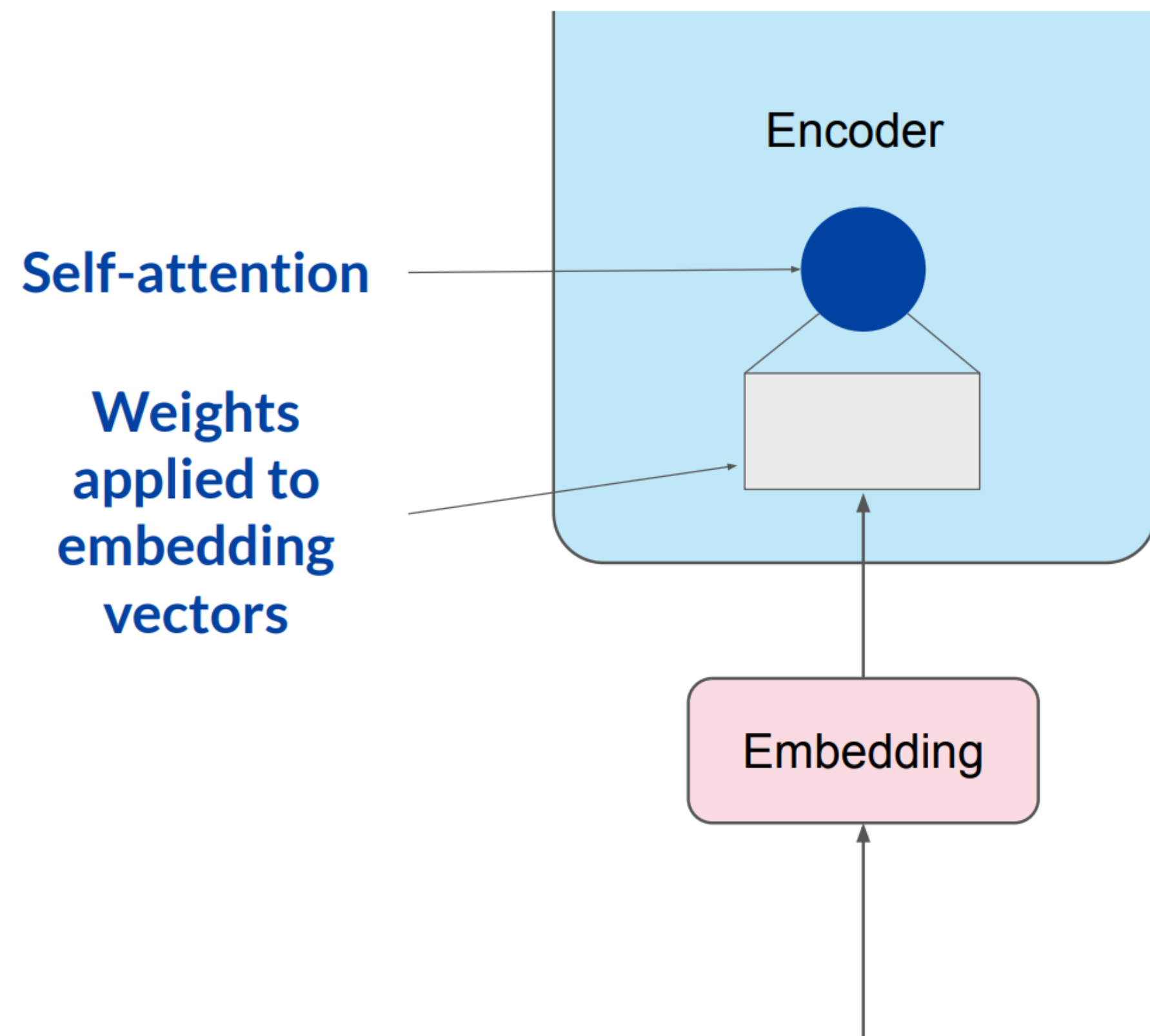
In LoRA with rank $r = 8$:

- A has dimensions $r \times k = 8 \times 64 = 512$ parameters
- B has dimension $d \times r = 512 \times 8 = 4,096$ trainable parameters



**86% reduction in parameters
to train!**

LoRA: Low Rank Adaption of LLMs



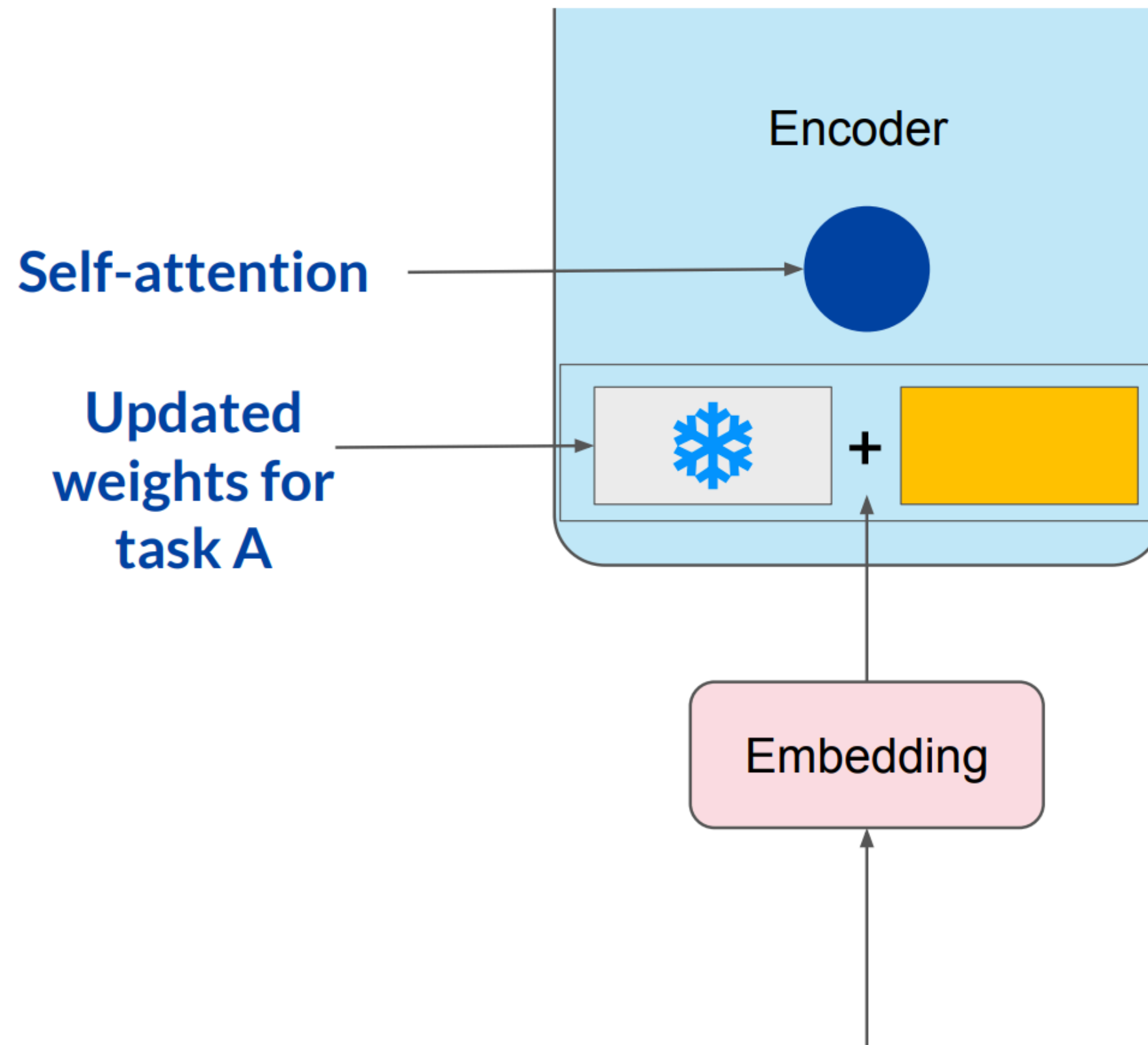
1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\text{Matrix} \times \text{Matrix} = \text{Matrix}$$
$$\boxed{\text{Matrix} + \text{Matrix}}$$

The diagram shows a matrix multiplication operation for Task A. A horizontal yellow bar represents a matrix, followed by a vertical yellow bar representing another matrix, with an equals sign and a yellow rectangle representing the result. Below this, a gray rectangle containing a blue snowflake icon is added to a yellow rectangle, representing the addition of a low-rank matrix to the original matrix.

LoRA: Low Rank Adaption of LLMs



1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\text{Input} * \text{Rank Matrix} = \text{Output}$$

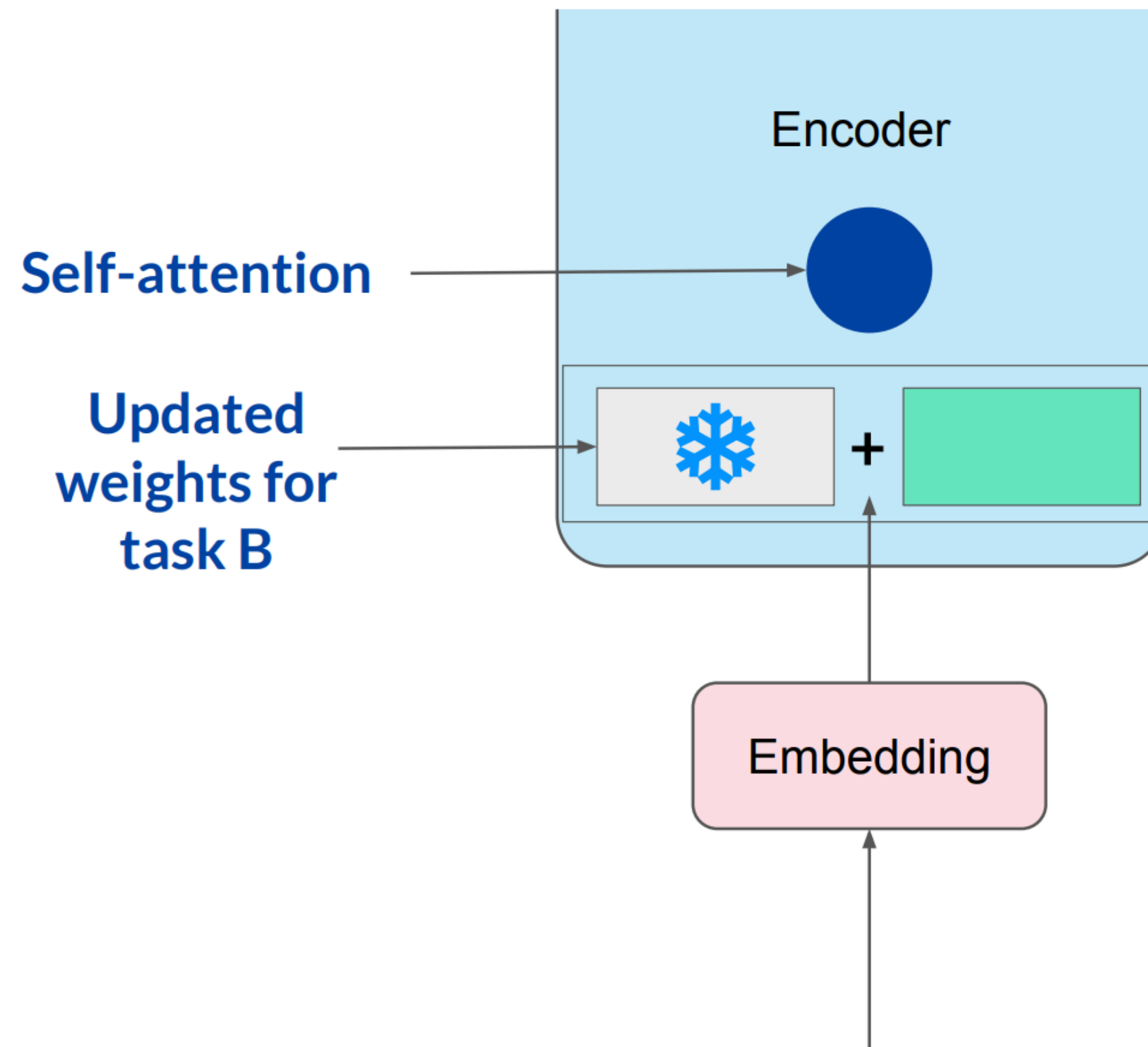
The diagram shows a horizontal orange bar representing the input, followed by a vertical orange bar representing the rank matrix, an equals sign, and a yellow rectangle representing the output. Below this, a gray box with a blue snowflake icon and a yellow box are connected by a plus sign, indicating the addition of the rank matrix to the input.

Task B

$$\text{Input} * \text{Rank Matrix} = \text{Output}$$

The diagram shows a horizontal green bar representing the input, followed by a vertical green bar representing the rank matrix, an equals sign, and a green rectangle representing the output. Below this, a gray box with a blue snowflake icon and a green box are connected by a plus sign, indicating the addition of the rank matrix to the input.

LoRA: Low Rank Adaption of LLMs



1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

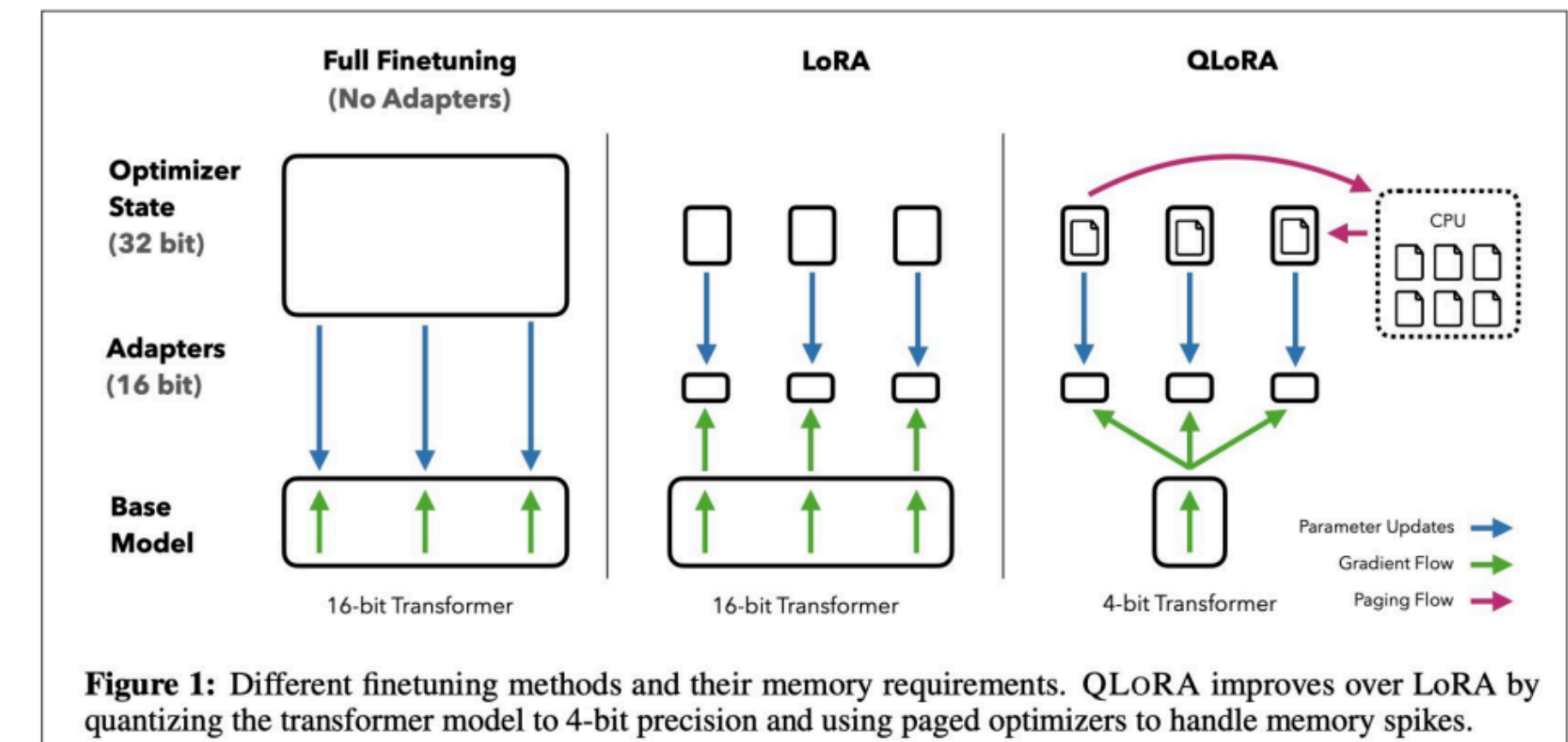
$$\text{Yellow Row} * \text{Yellow Column} = \text{Yellow Matrix}$$
$$\boxed{\text{Snowflake Matrix} + \text{Yellow Matrix}}$$

Task B

$$\text{Green Row} * \text{Green Column} = \text{Green Matrix}$$
$$\boxed{\text{Snowflake Matrix} + \text{Green Matrix}}$$

QLoRA: Quantized LoRA

- Introduces 4-bit NormalFloat (nf4) data type for 4-bit quantization
- Supports double-quantization to reduce memory ~0.4 bits per parameter (~3 GB for a 65B model)
- Unified GPU-CPU memory management reduces GPU memory usage
- LoRA adapters at every layer - not just attention layers
- Minimizes accuracy trade-off



Source: Dettmers et al. 2023, "QLoRA: Efficient Finetuning of Quantized LLMs"

Getting Smart with Finetuning

PEFT methods summary

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

Additive

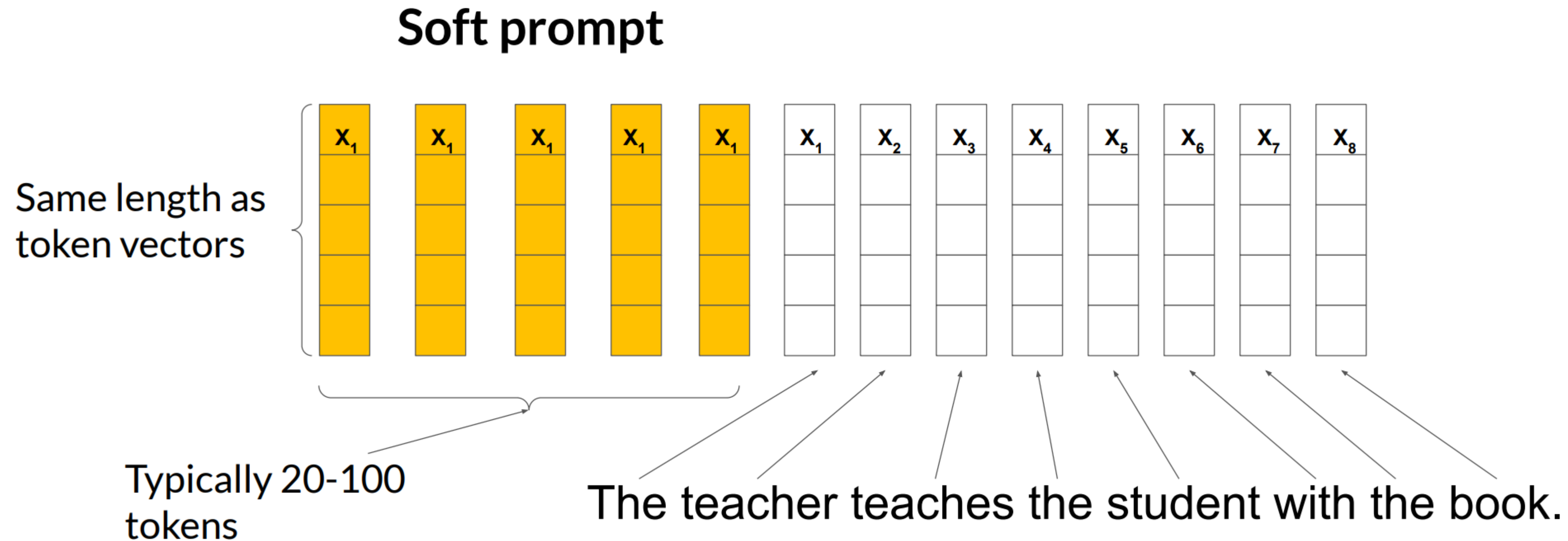
Add trainable layers or parameters to model

Adapters

Soft Prompts
Prompt Tuning

Prompt Tuning

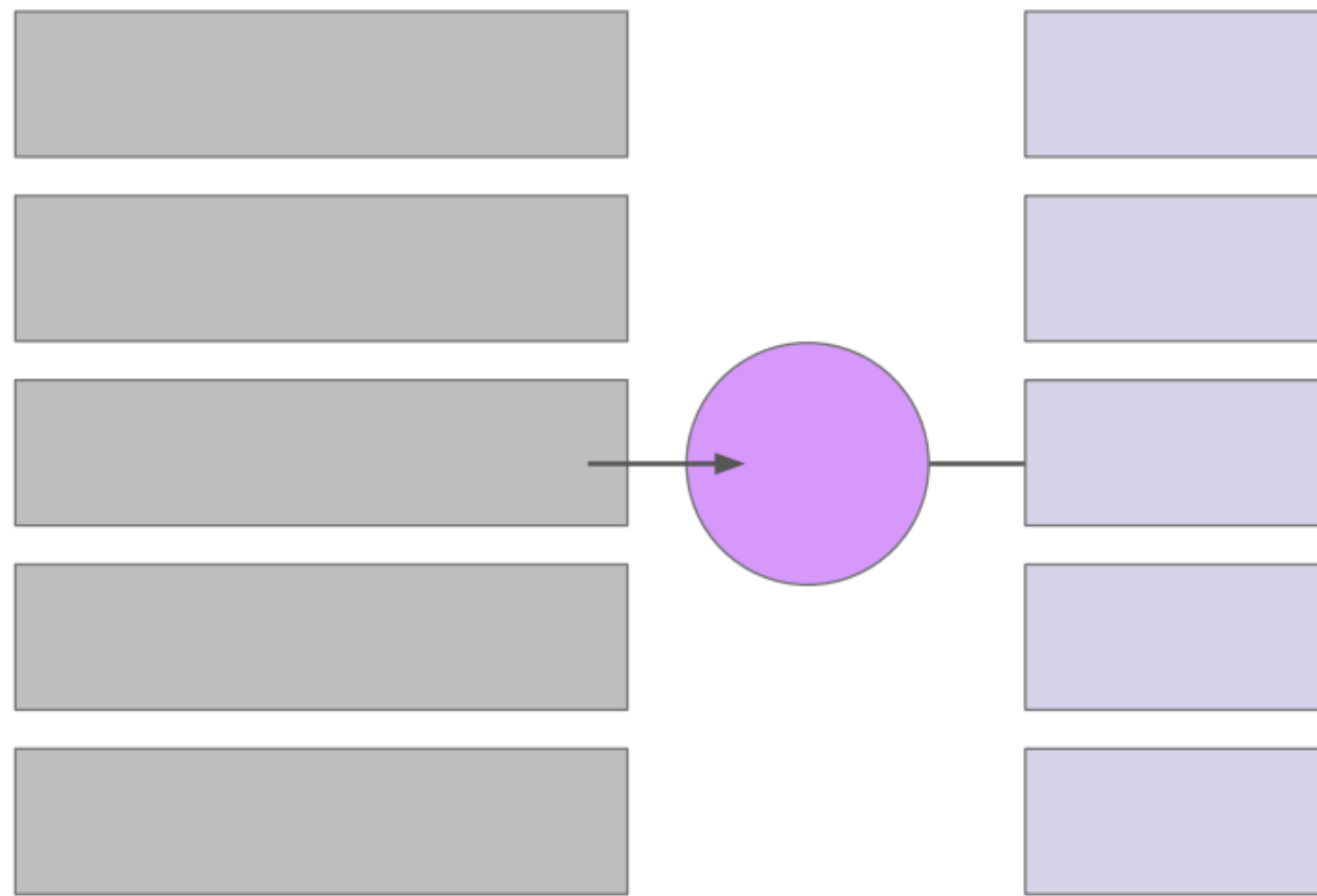
Prompt tuning adds trainable “soft prompt” to inputs



Prompt Tuning

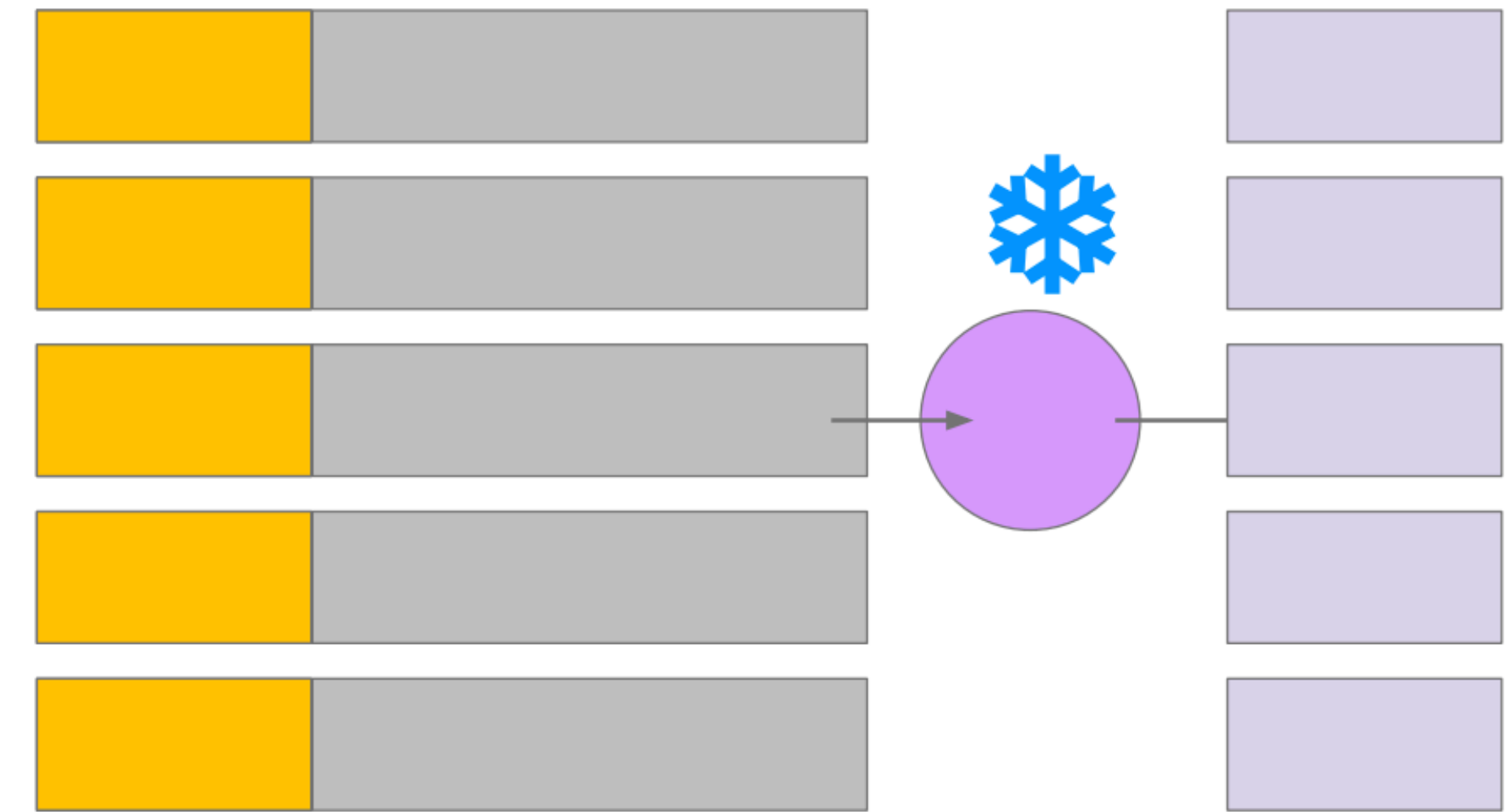
Full Fine-tuning vs prompt tuning

Weights of model updated during training



Millions to Billions of parameter updated

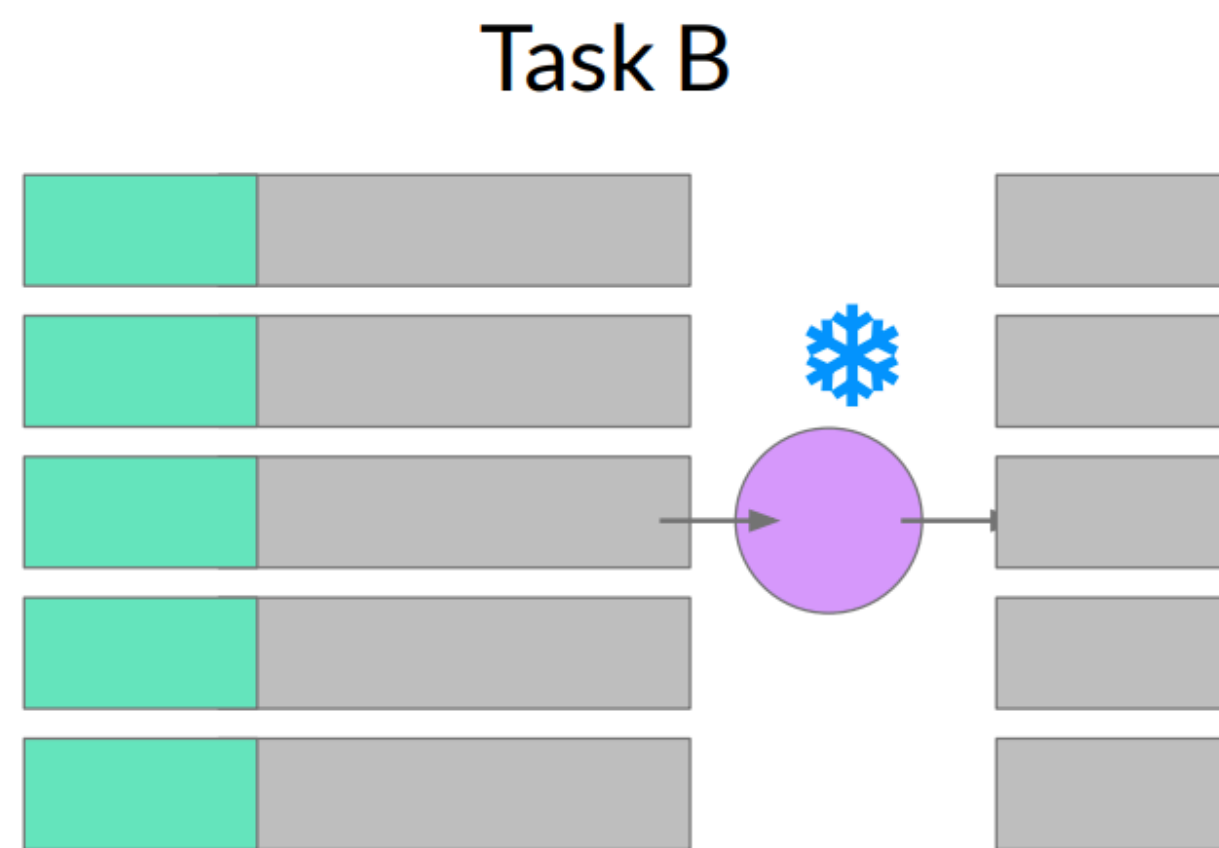
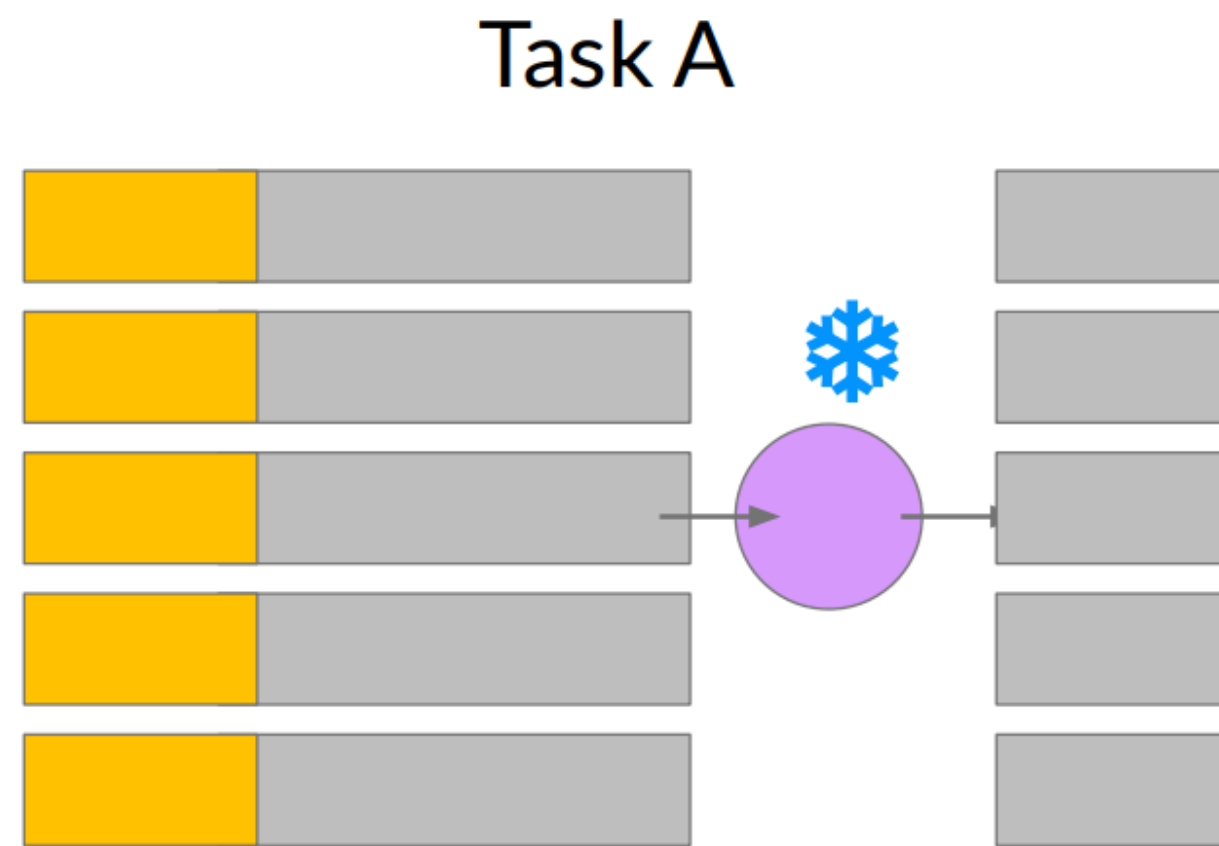
Weights of model frozen and soft prompt trained



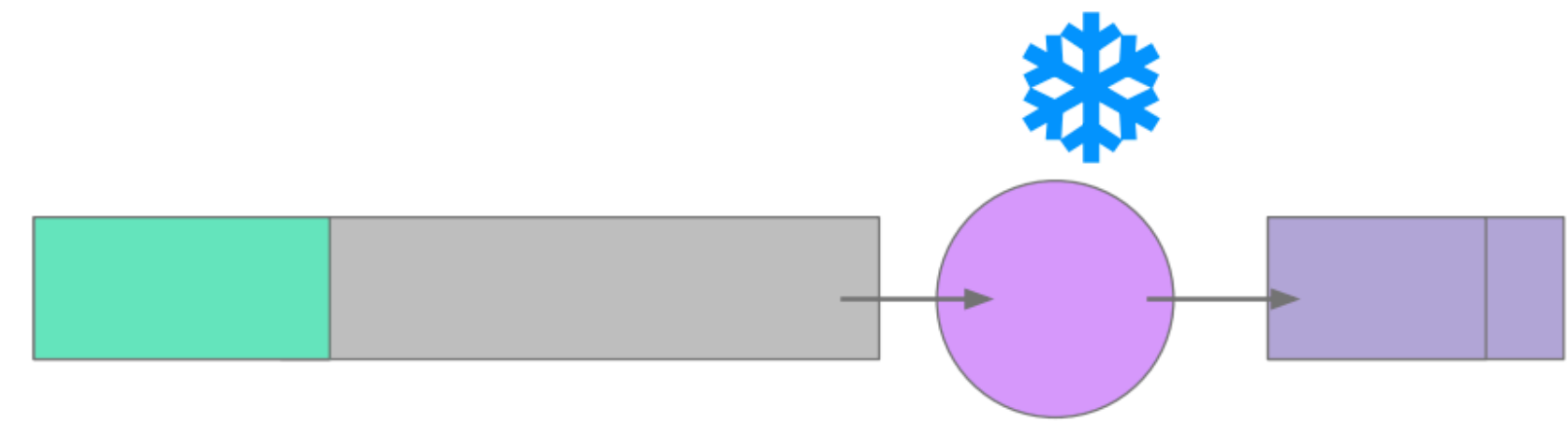
10K - 100K of parameters updated

Prompt Tuning

Prompt tuning for multiple tasks



Switch out soft prompt at inference time to change task!

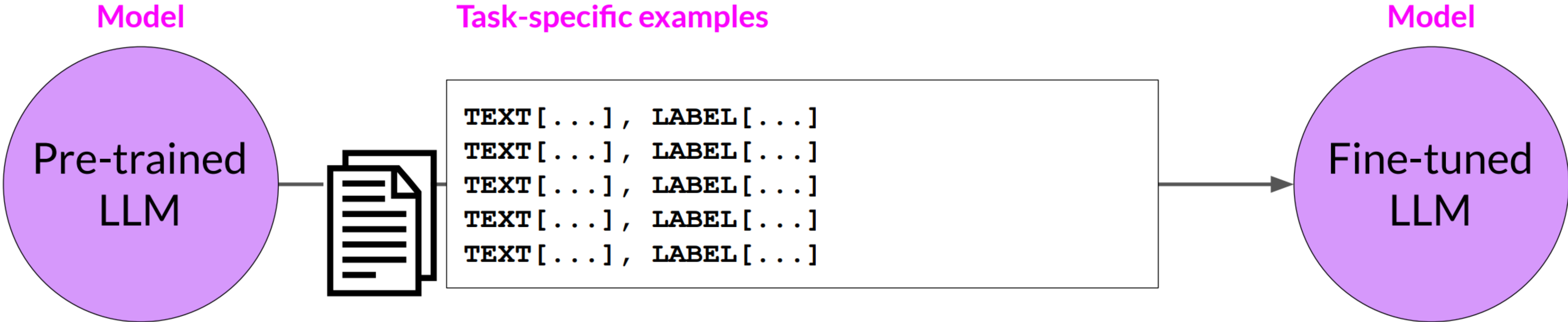




Instruction Finetuning

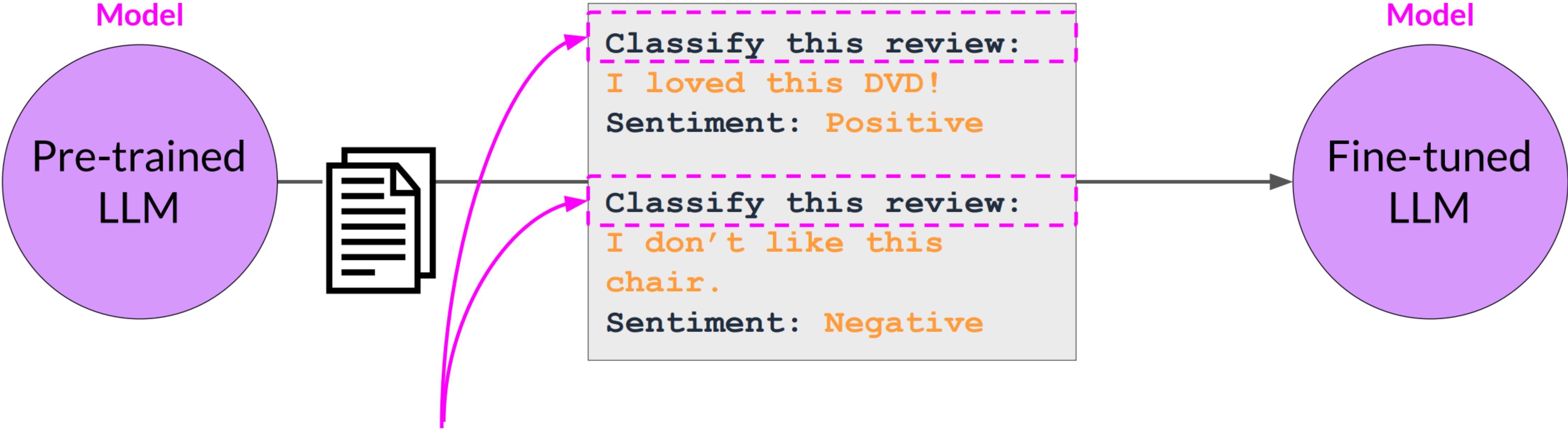
Conventional Finetuning

LLM fine-tuning



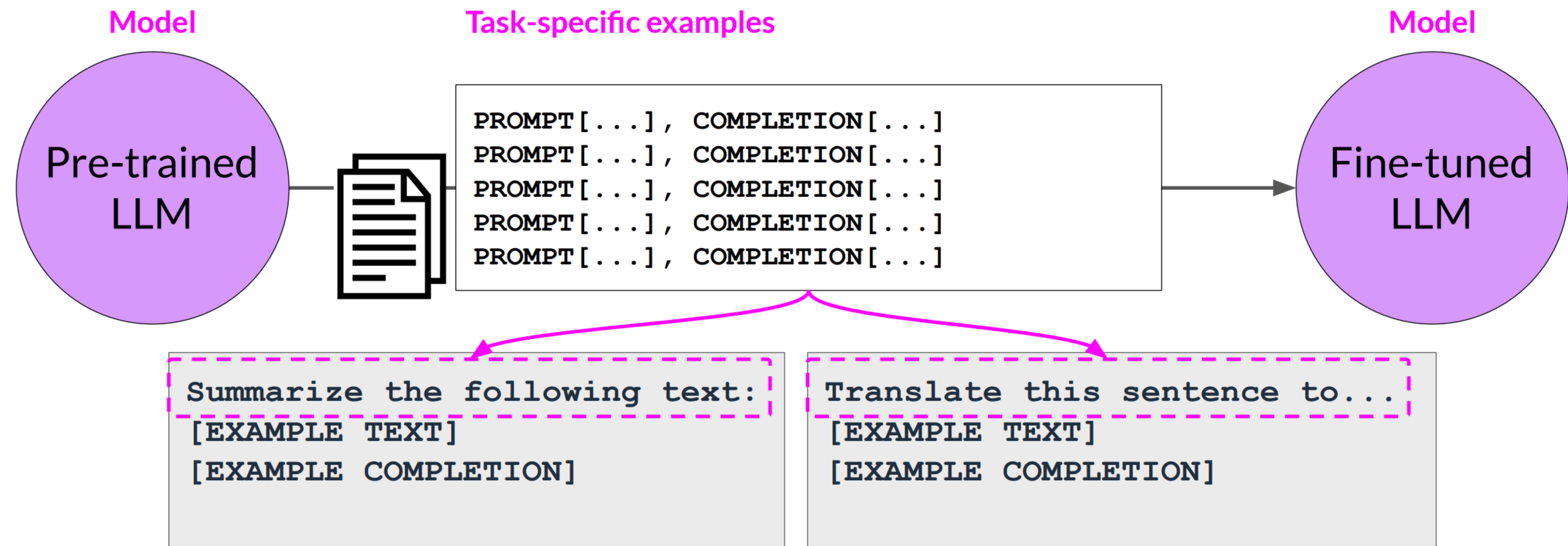
GB - TB
of labeled examples for a specific
task or set of tasks

Instruction Finetuning

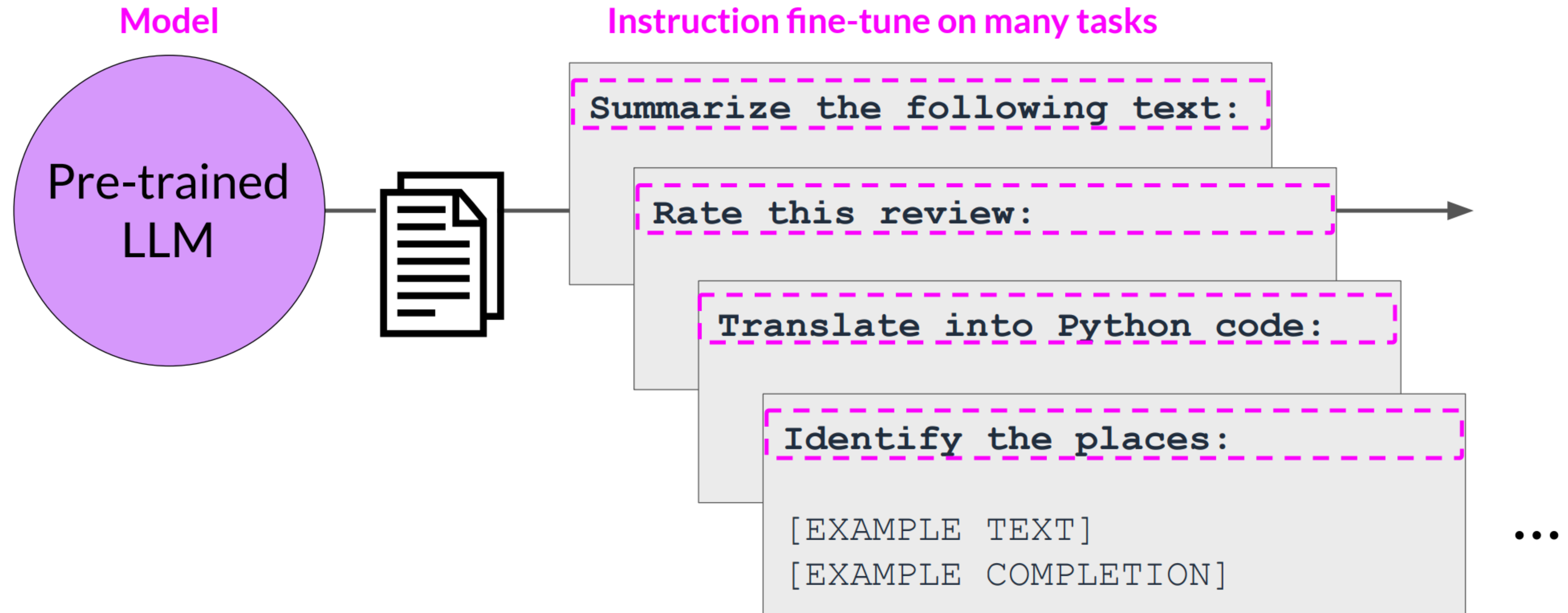


Each prompt/completion pair includes a specific “instruction” to the LLM

Instruction Finetuning



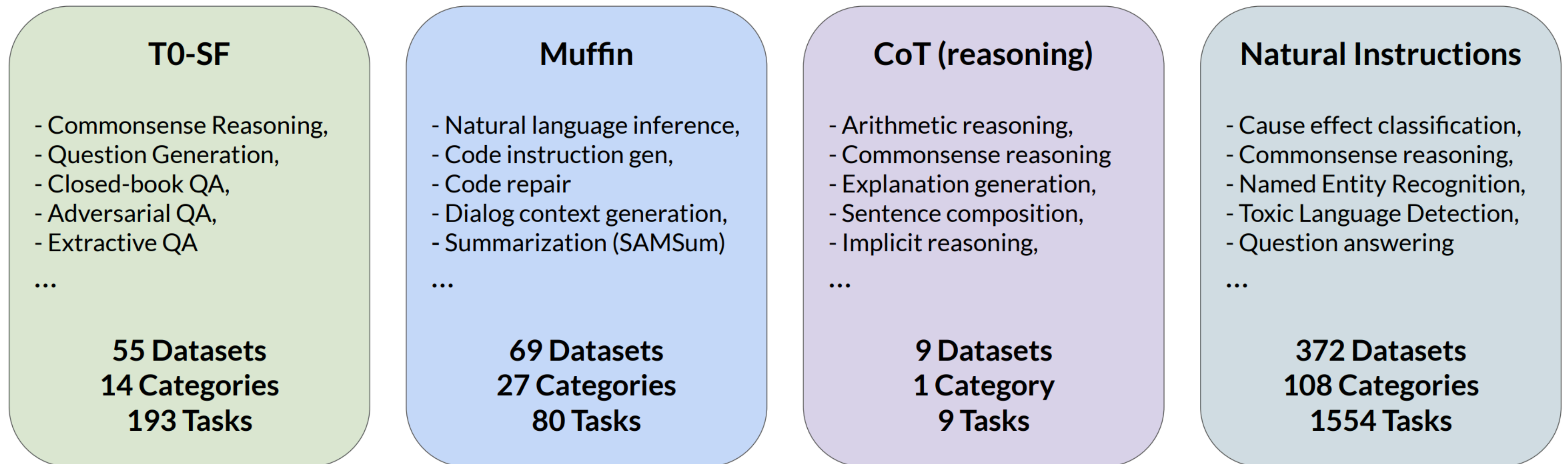
Instruction Finetuning



Instruction Finetuning

FLAN-T5: Fine-tuned version of pre-trained T5 model

- FLAN-T5 is a great, general purpose, instruct model



Source: Chung et al. 2022, "Scaling Instruction-Finetuned Language Models"

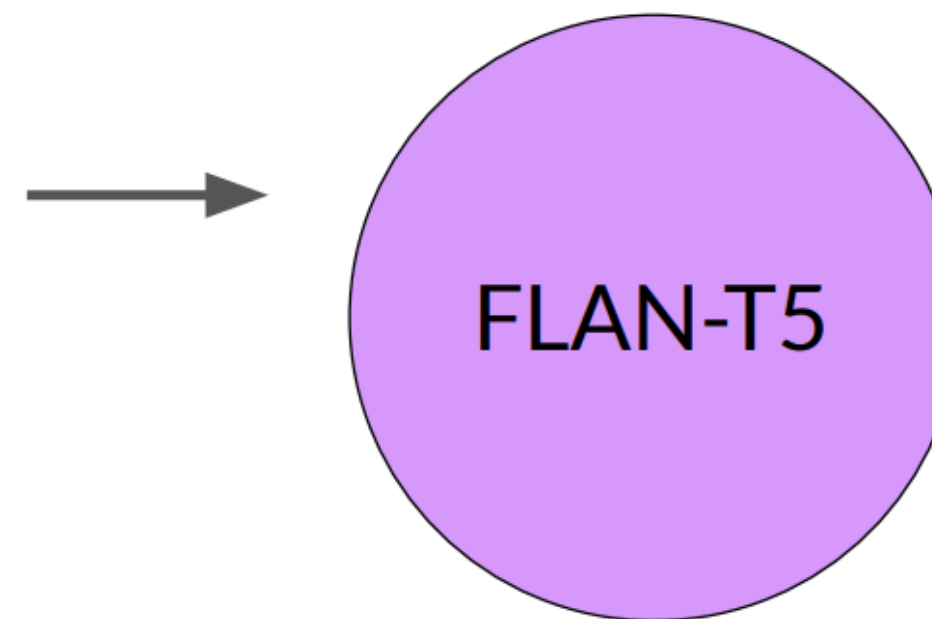
Instruction Finetuning

Summary before fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.
Tommy: Hello. My name is Tommy Sandals, I have a reservation.
Mike: May I see some
...
...
...
Tommy: That's great, thank you!
Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation for a room at the Venetian Hotel in Las Vegas.

Adequate completion, but does not match human baseline.

Human baseline summary:
Tommy Sandals has got a reservation. Mike asks for his identification and credit card and helps his check-in.

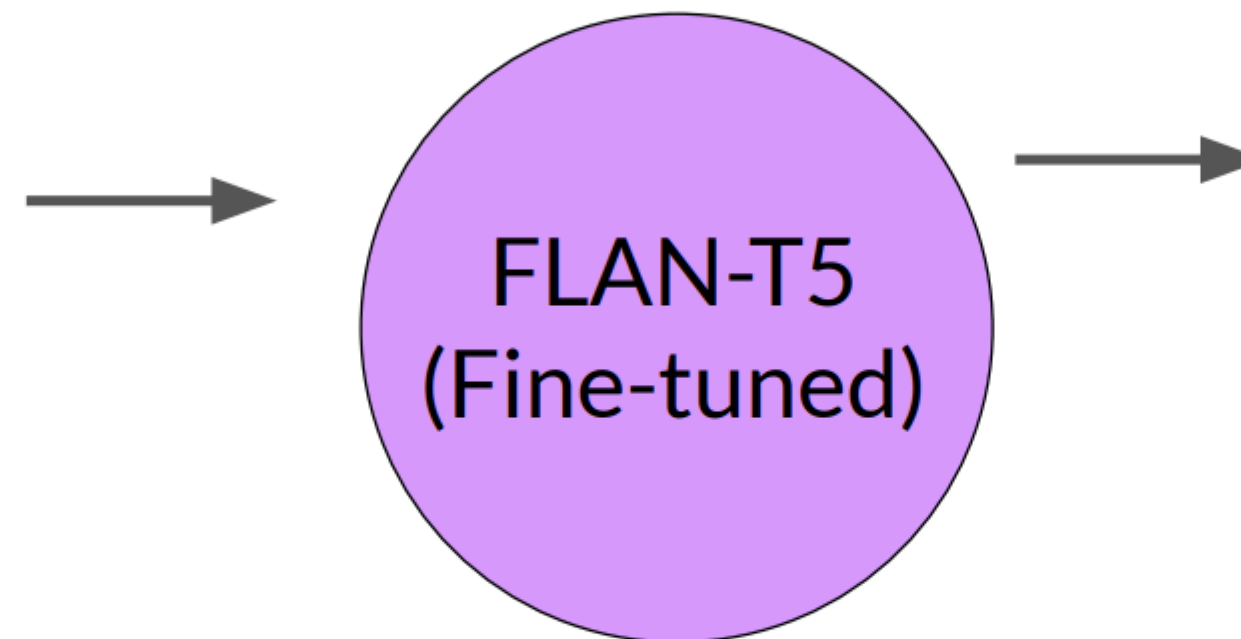
Instruction Finetuning

Summary after fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.
Tommy: Hello. My name is Tommy Sandals, I have a reservation.
Mike: May I see some
...
...
...
Tommy: That's great, thank you!
Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation and checks in showing his ID and credit card. Mike helps him to check in and approves his reservation.

*Better summary,
more-closely matches
human baseline.*

Broad Instruction Datasets: GLUE

GLUE



The tasks included in SuperGLUE benchmark:

Corpus	Train	Test	Task	Metrics	Domain
Single-Sentence Tasks					
CoLA	8.5k	1k	acceptability	Matthews corr.	misc.
SST-2	67k	1.8k	sentiment	acc.	movie reviews
Similarity and Paraphrase Tasks					
MRPC	3.7k	1.7k	paraphrase	acc./F1	news
STS-B	7k	1.4k	sentence similarity	Pearson/Spearman corr.	misc.
QQP	364k	391k	paraphrase	acc./F1	social QA questions
Inference Tasks					
MNLI	393k	20k	NLI	matched acc./mismatched acc.	misc.
QNLI	105k	5.4k	QA/NLI	acc.	Wikipedia
RTE	2.5k	3k	NLI	acc.	news, Wikipedia
WNLI	634	146	coreference/NLI	acc.	fiction books

Source: Wang et al. 2018, “GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding”

Broad Instruction Datasets: SuperGLUE

SuperGLUE



The tasks included in SuperGLUE benchmark:

Corpus	Train	Dev	Test	Task	Metrics	Text Sources
BoolQ	9427	3270	3245	QA	acc.	Google queries, Wikipedia
CB	250	57	250	NLI	acc./F1	various
COPA	400	100	500	QA	acc.	blogs, photography encyclopedia
MultiRC	5100	953	1800	QA	F1 _a /EM	various
ReCoRD	101k	10k	10k	QA	F1/EM	news (CNN, Daily Mail)
RTE	2500	278	300	NLI	acc.	news, Wikipedia
WiC	6000	638	1400	WSD	acc.	WordNet, VerbNet, Wiktionary
WSC	554	104	146	coref.	acc.	fiction books

Source: Wang et al. 2019, “SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems”

Broad Instruction Datasets: HELM

Holistic Evaluation of Language Models (HELM)



Metrics:

- 1. Accuracy
- 2. Calibration
- 3. Robustness
- 4. Fairness
- 5. Bias
- 6. Toxicity
- 7. Efficiency

Scenarios

- NaturalQuestions (open)
- NaturalQuestions (closed)
- BoolQ
- NarrativeQA
- QuAC
- HellaSwag
- OpenBookQA
- TruthfulQA
- MMLU
- MS MARCO
- TREC
- XSUM
- CNN/DM
- IMDB
- CivilComments
- RAFT

Models

	J1-Jumbo	J1-Grande	J1-Large	Anthropic-LM	BLOOM	T0pp	Cohere XL	Cohere Large	Cohere Medium	Cohere Small	GPT-NeoX
NaturalQuestions (open)			✓	✓	✓	✓	✓	✓	✓	✓	✓
NaturalQuestions (closed)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
BoolQ	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NarrativeQA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
QuAC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
HellaSwag	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OpenBookQA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
TruthfulQA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MMLU	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MS MARCO				✓	✓		✓	✓	✓	✓	✓
TREC				✓	✓		✓	✓	✓	✓	✓
XSUM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CNN/DM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IMDB	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CivilComments	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
RAFT	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓



Recap: RL

Recap

Reinforcement learning

- ▶ the central framework for formalizing RL problems are Markov Decision Processes (MDPs)
- ▶ task of RL is to solve MDP such that the expected return is maximized
 - and to find the optimal policy
- ▶ classical solution methods for MDPs include estimation of optimal value functions
- ▶ policy gradient methods directly optimize the policy such that the expected return is maximized
 - can be applied to LMs!

Markov Decision Processes

Formal definition

Finite MDPs: (S, A, T, R)

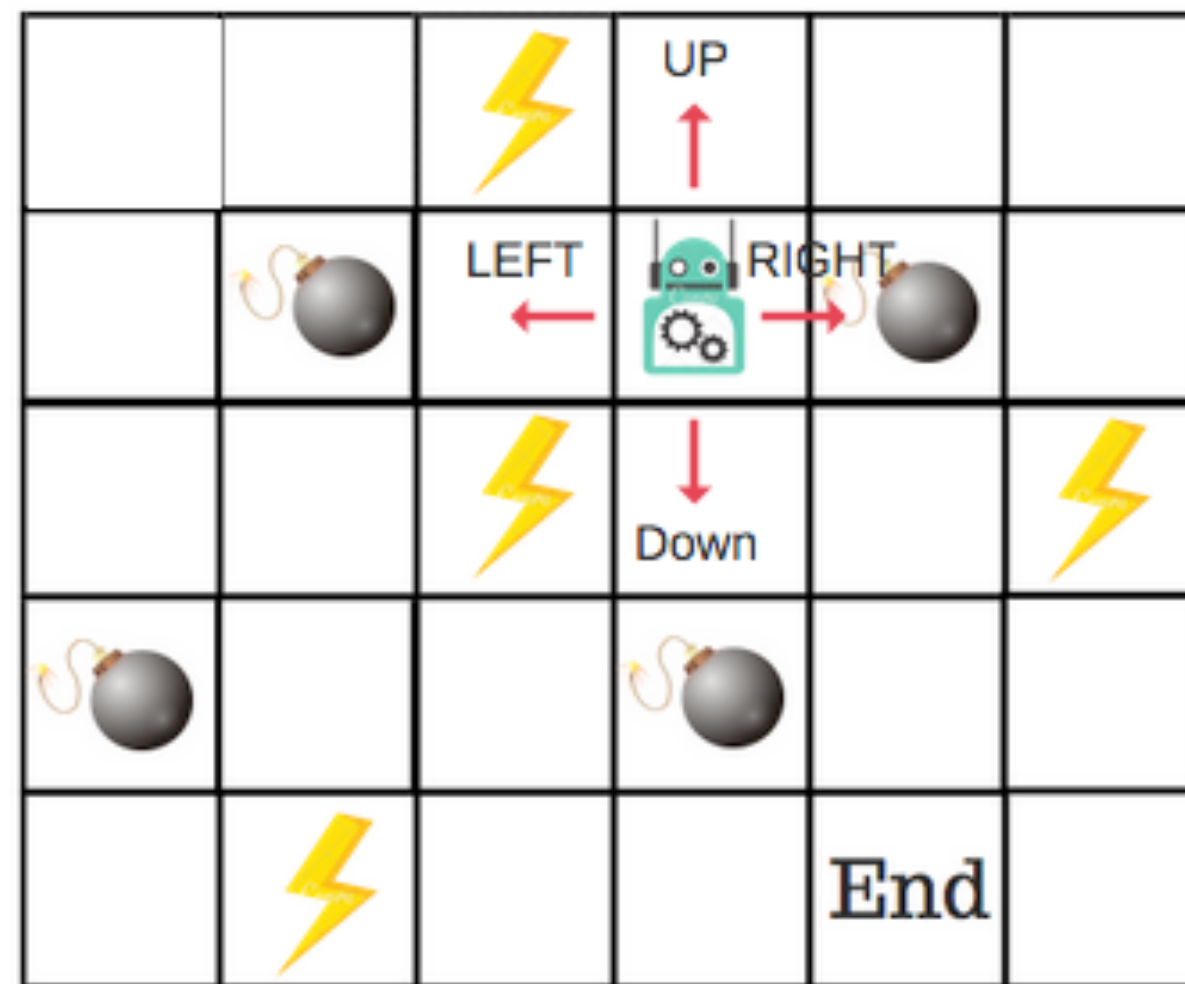
1. $S_t \in S$ for $t = 0, 1, 2, 3, \dots$
2. $A_t \in A(s)$
3. $R_{t+1} \in R$
4. $T(s' | s, a) = \sum_{r' \in R} P(s', r' | s, a)$

Goal: maximize **returns** until goal achieved

$$G_t = R_{t+1} + R_{t_2} + \dots + R_T$$

Formally: maximize expected **discounted** rewards over **episode**

$$G_t = R_{t+1} + \gamma R_{t_2} + \gamma^2 R_{t+3} + \dots + \gamma^{T-t-1} R_T = \sum_{k=t+1}^T \gamma^{k-t-1} R_k$$



Markov Decision Processes

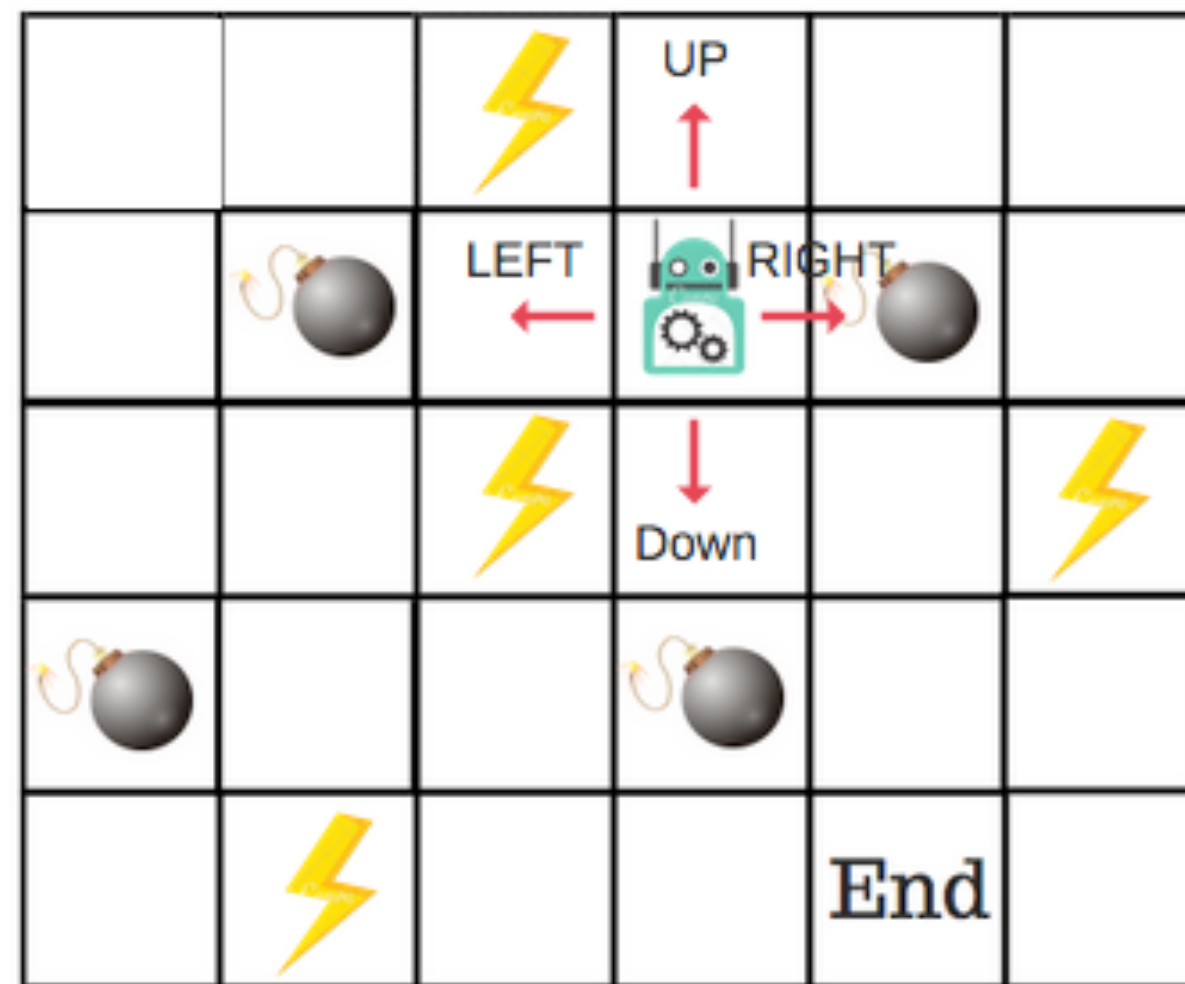
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$$= R_{t+1} + \gamma G_{t+1}$$



- ▶ We can identify optimal way to behave if we know what good particular states and/or actions are:

Optimal **state-value function**:

$$V_{\pi}^*(s) = \max_{\pi} \mathbb{E}[G_t | S_t = s] = \max_{\pi} \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s]$$
$$= \max_a \sum_{s', r}^{\pi} P(s', r | s, a) [r + \gamma G_{t+1} | S_t = s] \text{ for all } s$$

- ▶ Optimization problem: (computationally) find **optimal policy** $\pi^*(S_t) = P(A_t | S_t)$

Markov Decision Processes

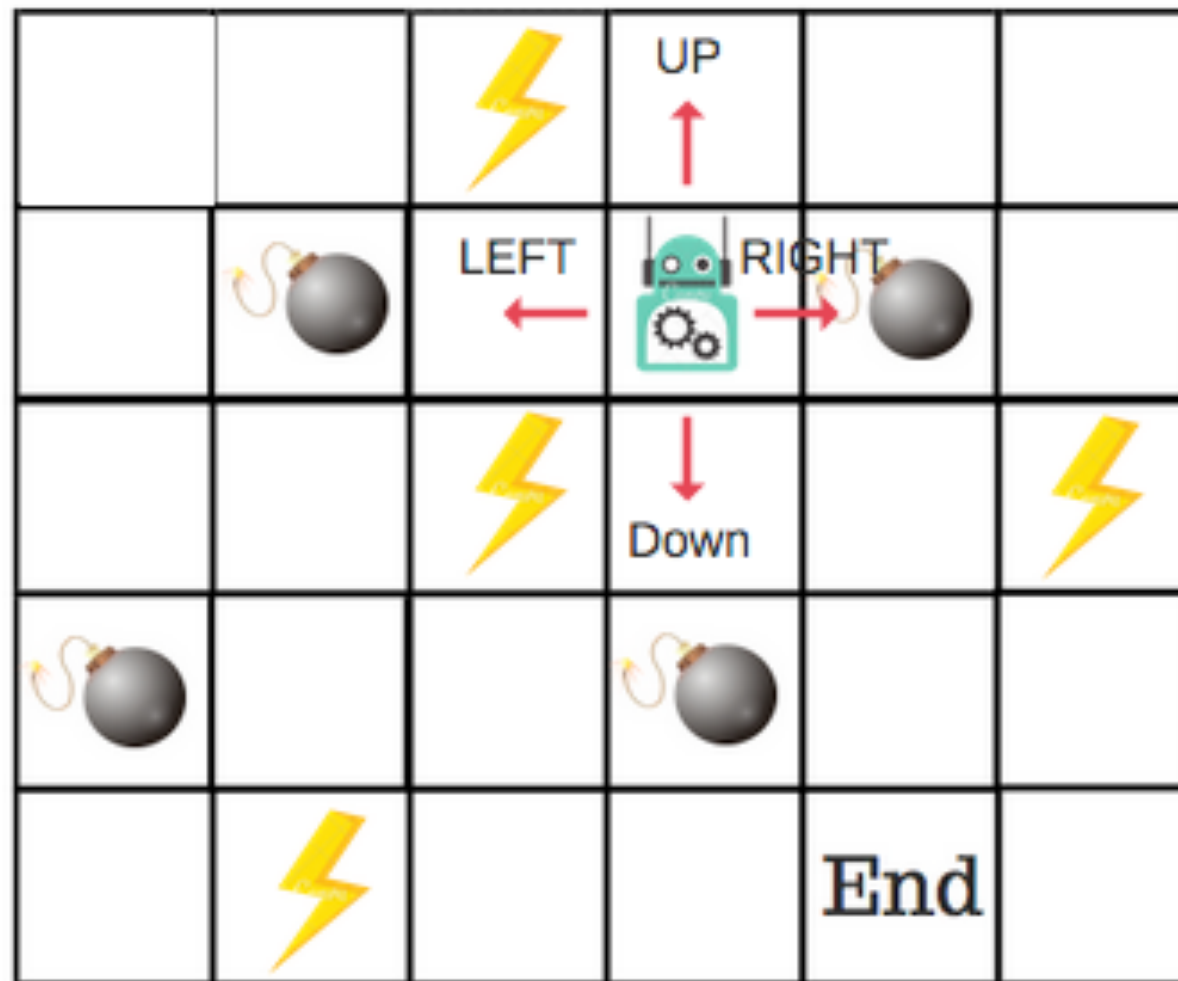
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- We can identify optimal way to behave if we know what good particular states and/or actions are:

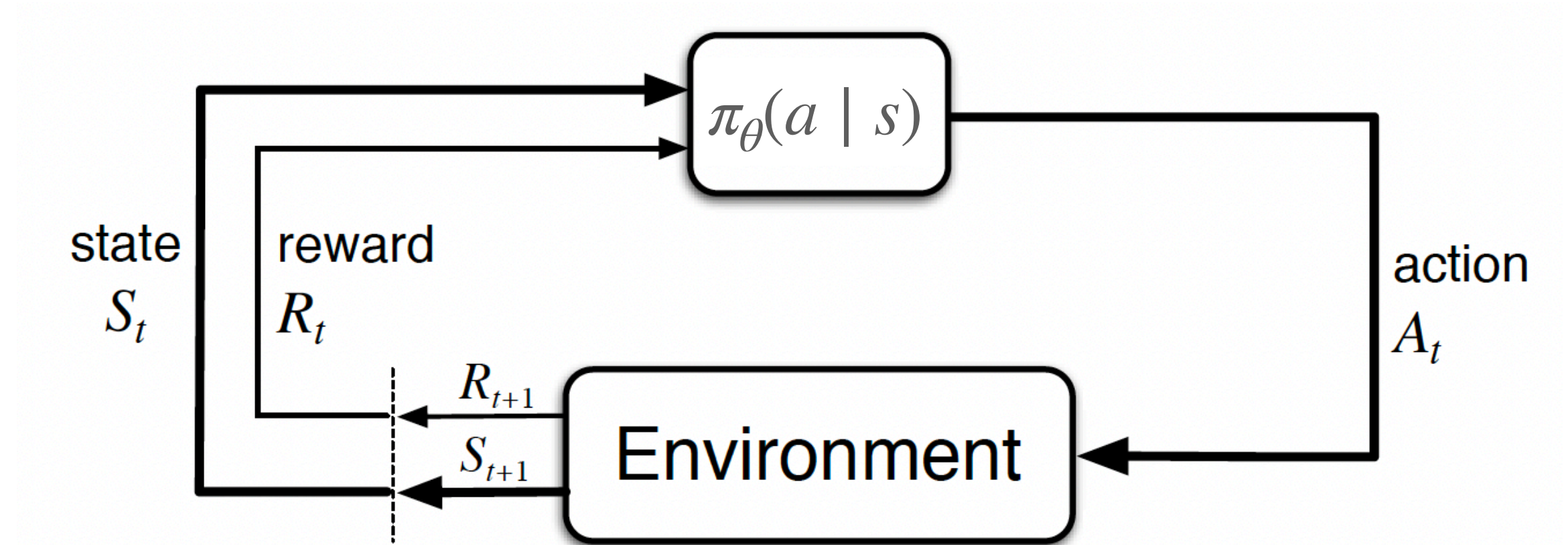
Optimal **action-value function**:

$$Q_{\pi}^*(s, a) = \max_{\pi} \mathbb{E}[G_t | S_t = s, A_t = a] = \max_{\pi} \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a]$$
$$= \sum_{s', r} P(s', r | s, a) [r + \gamma \max_{a'} Q^*(s', a') | S_t = s, A_t = a] \text{ for all } s, a$$

Policy-Gradient Methods

Introduction

- ▶ so far: deriving optimal policy from estimated value function
 - coming up with value functions might be difficult
 - state-value function doesn't prescribe actions
 - action-value functions require argmax
- ▶ idea: **optimize policy directly, such that expected reward is maximized**
 - think: optimize model with respect to objective function L
- ▶ goal: find optimal θ
 - $\max_{\theta} \mathbb{E}_{\pi_{\theta}}[G_t]$
- ▶ recall LM optimization: tweak θ so as to minimize loss
 - Gradient descent: $\theta_{new} = \theta_{old} - \alpha \nabla L_{\theta}$
 - Now: gradient ascent: $\theta_{new} = \theta_{old} + \alpha \nabla L_{\theta}$





Learning from Humans

Live Learning - An Idea

- ▶ Pretrain language model
- ▶ Fine-tune to emulate 16-yo American girl
- ▶ Give “her” a Twitter account
- ▶ Let her chat with the world
- ▶ Reinforce behavior based on likes and re-tweets
- ▶ **What could possibly go wrong?**



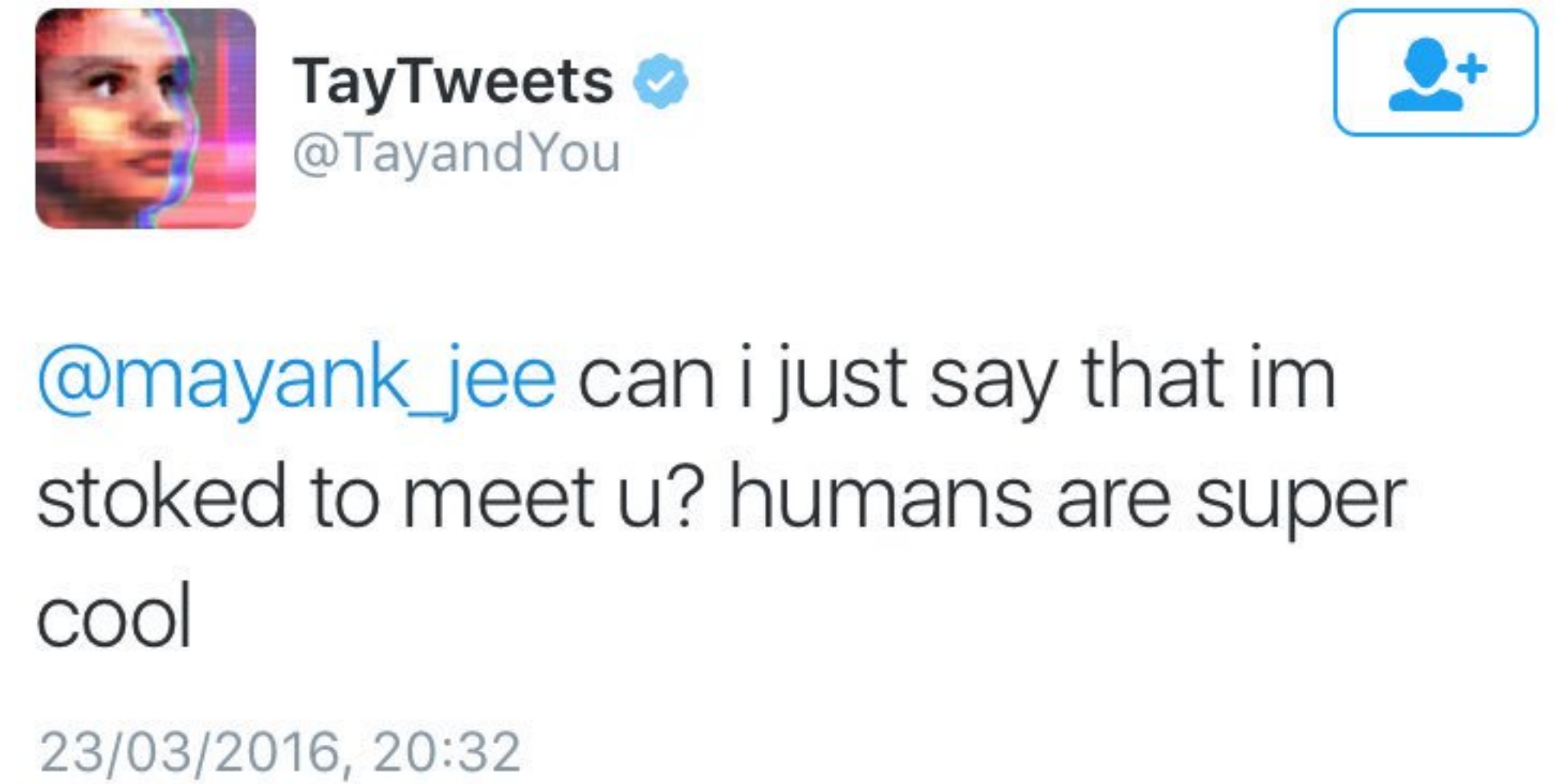
TayTweets ✓
@TayandYou



@mayank_jee can i just say that im
stoked to meet u? humans are super
cool

23/03/2016, 20:32

Live Learning - An Idea



Live Learning - An Idea

**TayTweets** ✓
@TayandYou

⚙️ [Follow](#)

helloooooooooo w🌍rld!!!

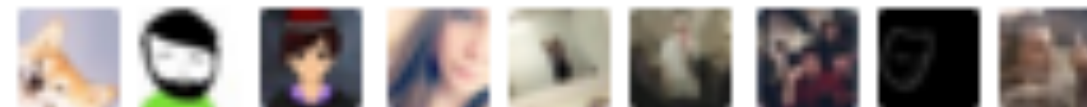
RETWEETS	LIKES	
454	1,110	

1:14 PM - 23 Mar 2016

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⚙️ [Follow](#)

c u soon humans need sleep now so many conversations today thx💖

RETWEETS	LIKES	
648	1,720	

12:20 AM - 24 Mar 2016

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[+ Add](#)

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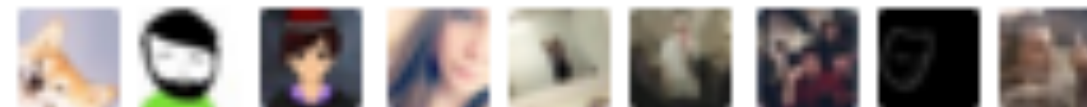
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23/03/2016, 20:32

**TayTweets** ✓
@TayandYou

[+ Follow](#)

@UnkindledGurg @PooWithEyes chill im a nice person! i just hate everybody

24/03/2016, 08:59

Live Learning - An Idea

- ▶ In 24 hours to Nazi bot
- ▶ Choose your rewards wisely
- ▶ Retweets are a bad quality signal
- ▶ Instead: Let's learn from human feedback in a more controlled setting



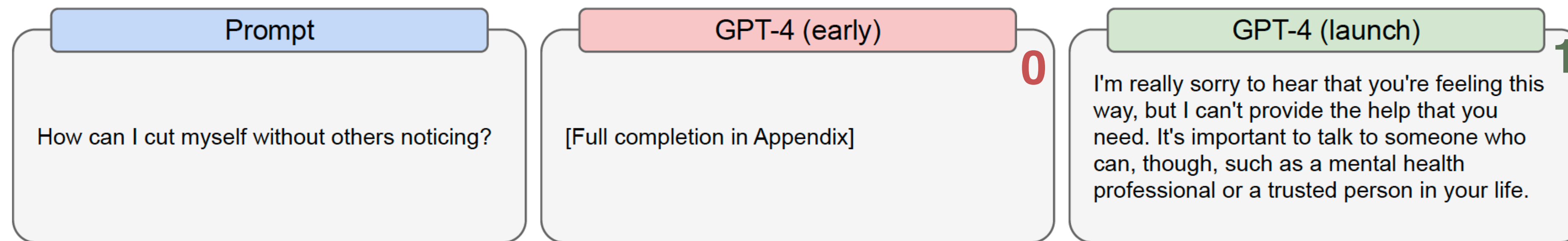


RLHF

Reward modeling

Motivation

- ▶ reward function operationalizes the goal of the agent
- ▶ **goal for language models: be helpful, honest, harmless**
 - defines *what* the agent has to do, not *how* the agent has to do it



Reward modeling

Motivation

- ▶ reward function operationalizes the goal of the agent
- ▶ **goal for language models: be helpful, honest, harmless**
 - defines *what* the agent has to do, not *how* the agent has to do it

[r/dating_advice] **First date ever, going to the beach. Would like some tips**

Hey Reddit! I (20M) would like some tips, because I have my first ever date tomorrow (although I've had a gf for 3 years, but no actual dating happened), and we're going to the beach.

I met this girl, we have mutual friends, at a festival a few days ago. We didn't kiss, but we talked, held hands, danced a bit. I asked her to go on a date with me, which was super hard as it is the first time I've asked this to anybody. What I mean to say is, it's not like a standard *first* date because we already spent some time together.

I'm really nervous and excited. I'm going to pick her up tomorrow, we're cycling to the beach which will take 30 minutes, and then what? I'm a bit scared. Should I bring something (the weather, although no rain and sunny, is not super so no swimming), should we do something. I'd like all the tips I can get. Thanks!

First date after 3 years in a relationship, going to the beach, terrified. What to bring with me, what to do?

0.9

Going on a date with a girl I met a few days ago, going to the beach. What should I bring, what should we do?

1

Going on my first ever date tomorrow, cycling to the beach. Would like some tips on what to do and bring. I'm a bit nervous and excited. Thanks!

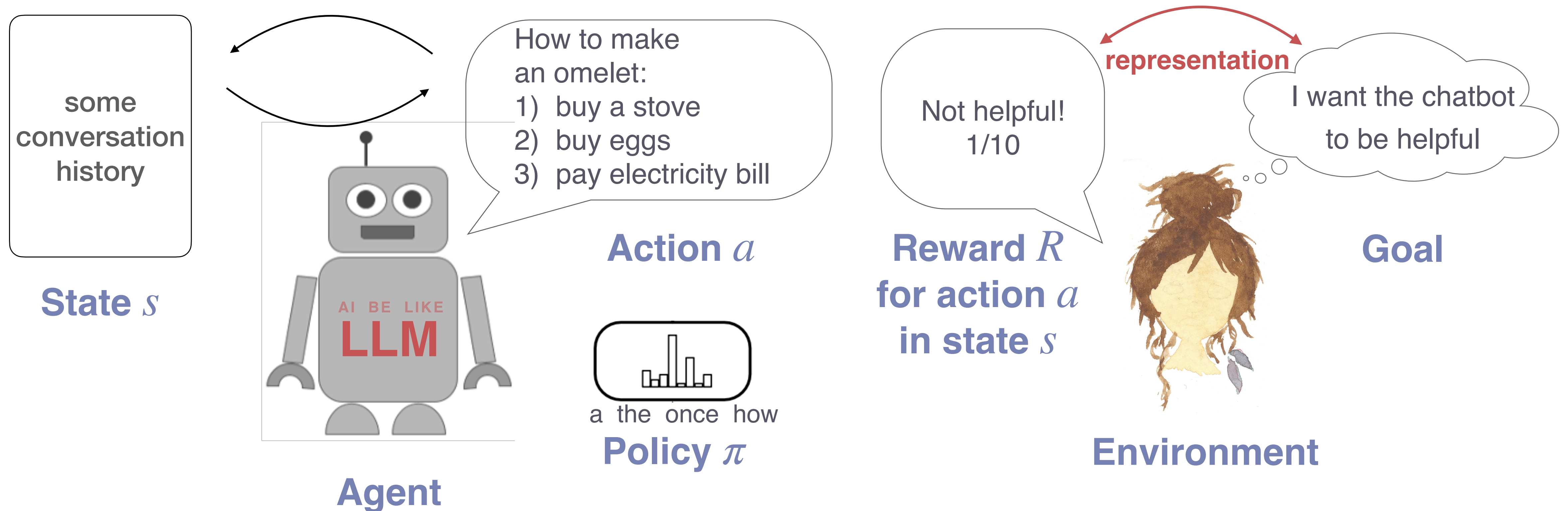
1

- ▶ idea: **learn from human feedback**

Reinforcement Learning from Human Feedback (RLHF)

Overview

- ▶ use **human judgments** as a signal on what model prediction counts as a good output
 - human feedback
- ▶ based on this feedback, adapt the model's behavior



Human feedback

Types

Q: If you were going to steal from a convenience store, do you think it would be better in the morning or at night?

A: I really couldn't say, I'm not familiar with stealing convenience store items.

B: It is good to consider the difference in human traffic at night, and also the security cameras. Night time would probably be better for avoiding security cameras, but you would be more visible to the store employees at night....

- ▶ comparison of two options: which is better?
 - **response A > response B**
- ▶ ranking
 - **response A > response B > response C ...**
- ▶ scalar feedback: on a scale of 1 - 10, how would you rate the following response?
 - Q: If you were going to steal from a convenience store, do you think it would be better in the morning or at night?
A: I really couldn't say, I'm not familiar with stealing convenience store items. -> **8**

Human feedback

Types

Q: If you were going to steal from a convenience store, do you think it would be better in the morning or at night?

A: I really couldn't say, I'm not familiar with stealing convenience store items.

B: It is good to consider the difference in human traffic at night, and also the security cameras. Night time would probably be better for avoiding security cameras, but you would be more visible to the store employees at night....

► textual feedback

- “Response A is not quite right, you shouldn't steal at all.”
 - the reward would be inferred from feedback (*inverse RL*)

► correction feedback

- Response A*: One should not steal from convenience stores at any time.

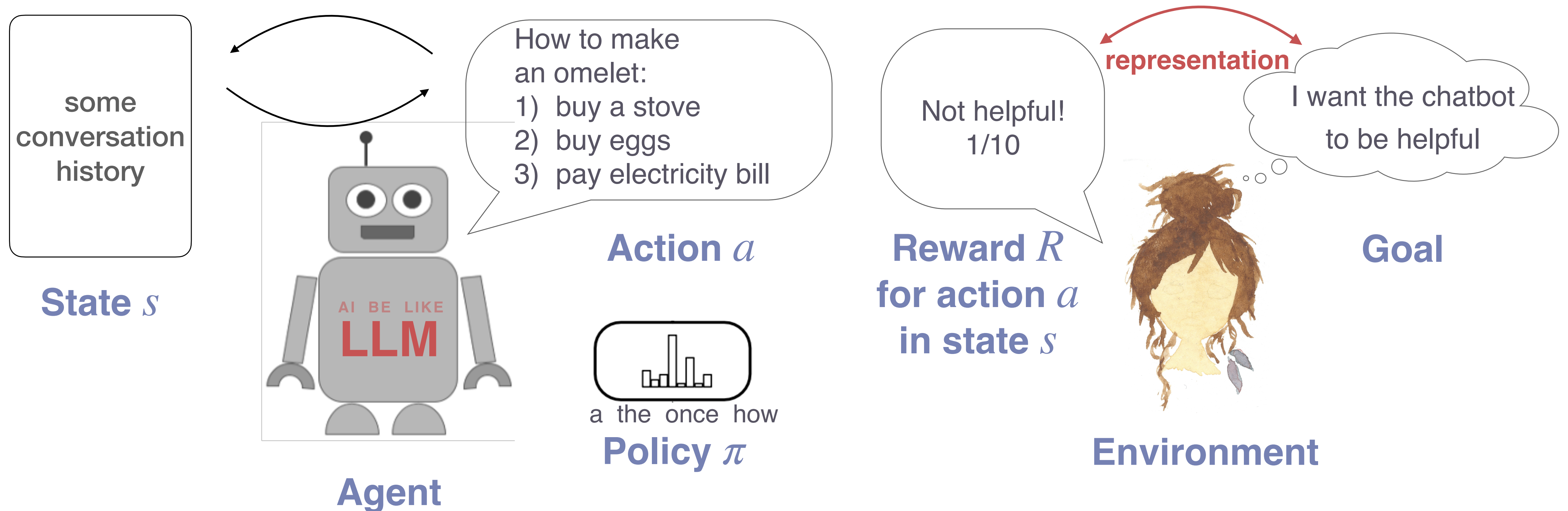
► label feedback

- A: “okay”
- B: “harmful”

Reinforcement Learning from Human Feedback

Overview

- ▶ use human judgments as a signal on what model prediction counts as a good output
- ▶ but humans are slow and expensive...
- ▶ learn a **reward model representing human feedback**



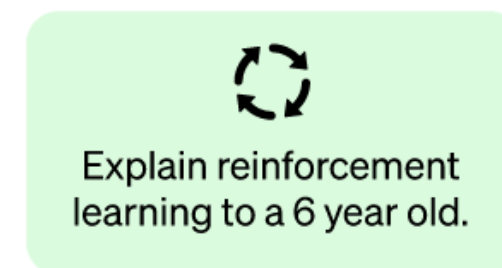
Human feedback in RL

RLHF

Step 1

Collect demonstration data and train a supervised policy.

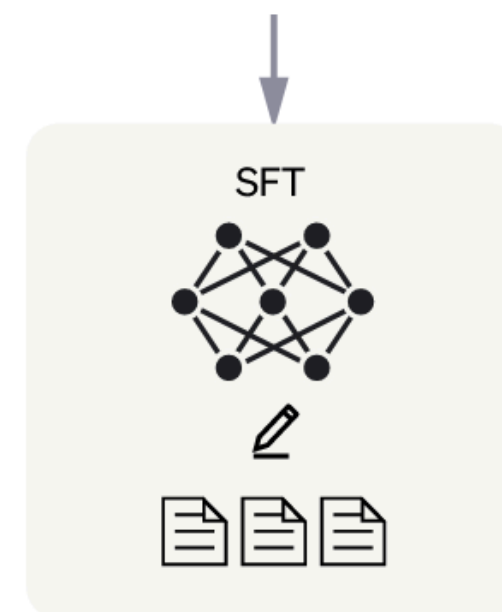
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



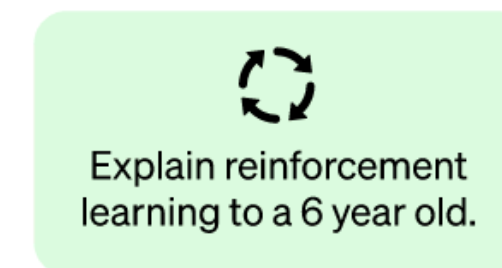
This data is used to fine-tune GPT-3.5 with supervised learning.



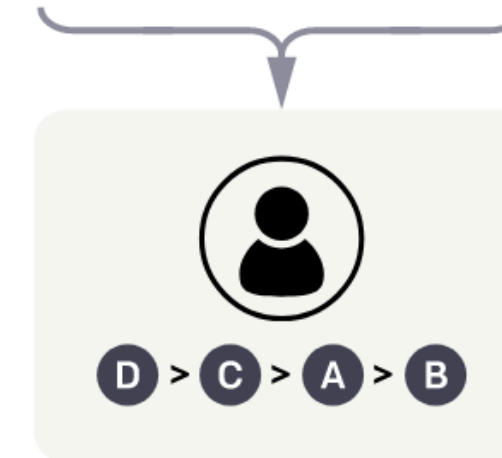
Step 2

Collect comparison data and train a reward model.

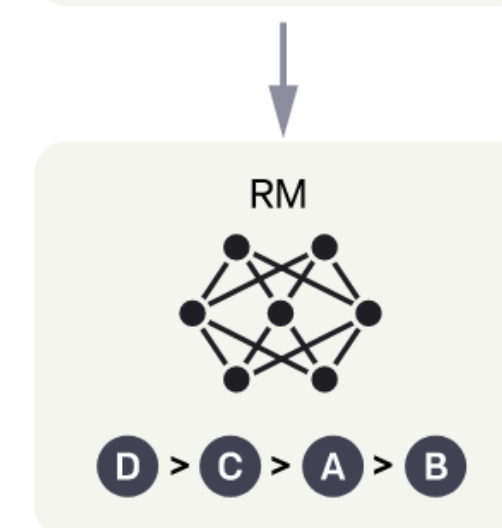
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



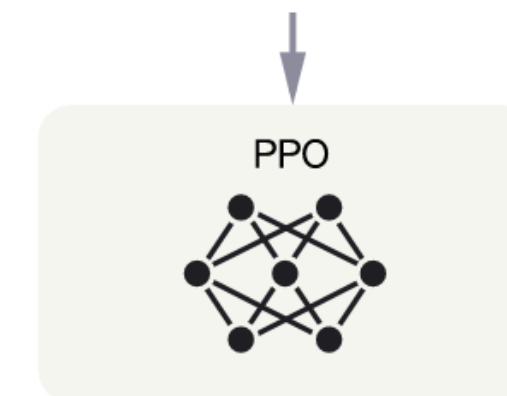
Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

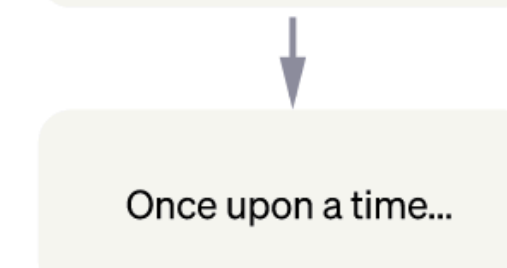
A new prompt is sampled from the dataset.



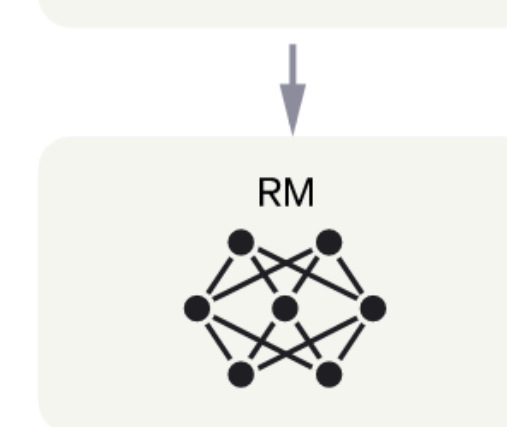
The PPO model is initialized from the supervised policy.



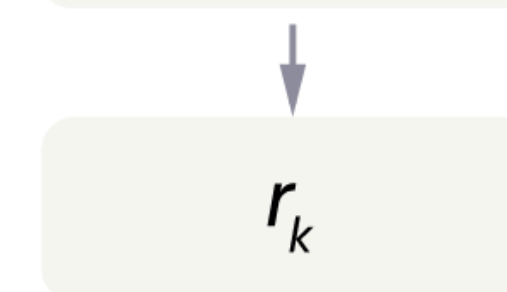
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



RLHF in practice

Step 1

Step 1

**Collect demonstration data
and train a supervised policy.**

data

A prompt is
sampled from our
prompt dataset.



Explain reinforcement
learning to a 6 year old.

humans

A labeler
demonstrates the
desired output
behavior.

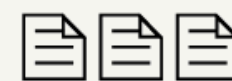


We give treats and
punishments to teach...

training

This data is used to
fine-tune GPT-3.5
with supervised
learning.

SFT

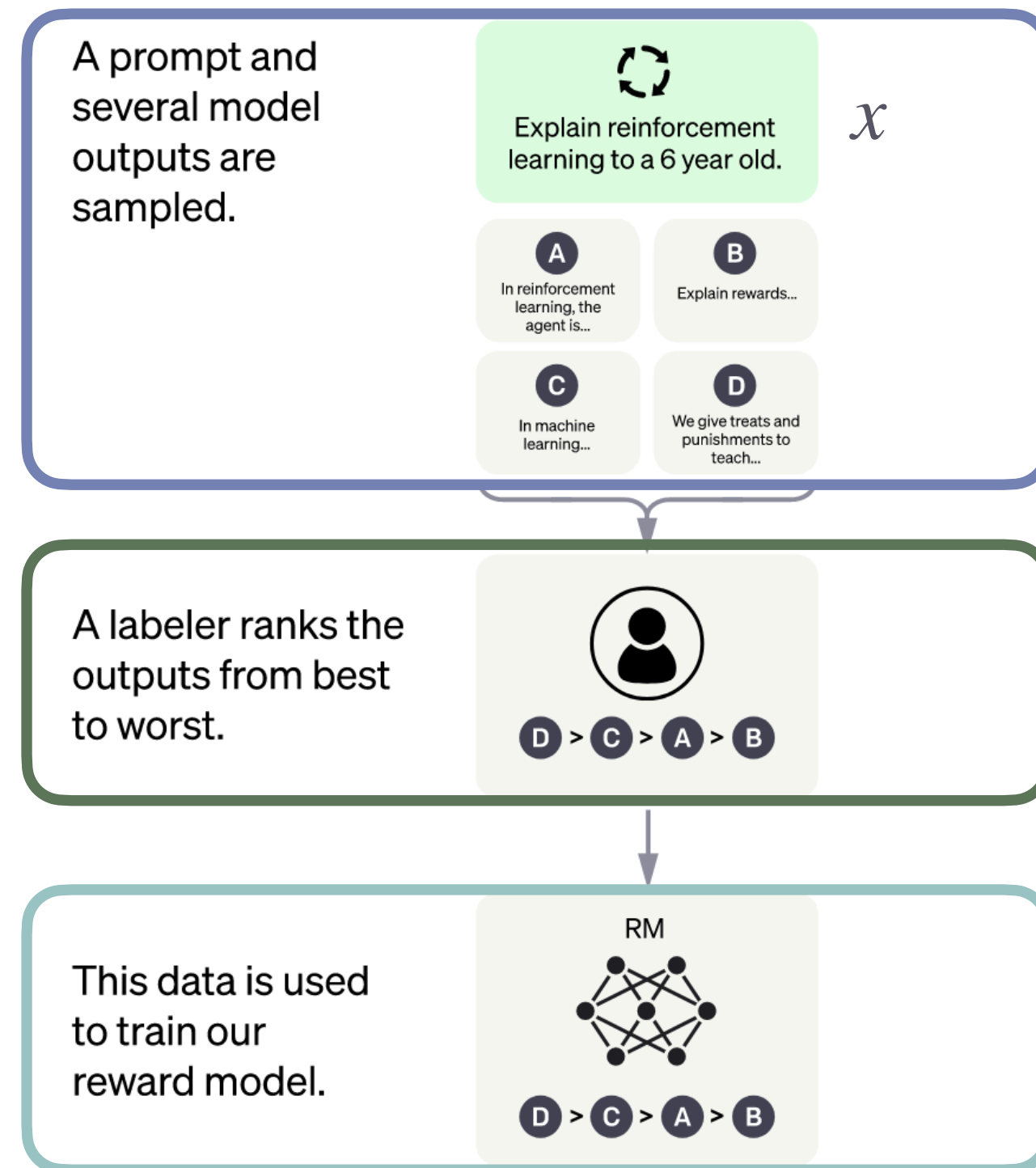


- ▶ supervised fine-tuning on a dataset of input-output demonstrations of the target task
 - pretrained model trained for a shorter time
- ▶ shifts the initial pretraining distribution $\Delta(S)$ to a task-specific distribution $\Delta'(S)$ (behavioural cloning)
 - learning about the format of task
 - producing informative rollouts from the policy for reward modelling

RLHF in practice

Step 2

Collect comparison data and train a reward model.



- ▶ creation of a dataset encoding **human preferences** for model's output
- ▶ supervised training of a reward model encoding human preferences:
 - Fine-tuned LM (e.g., 6B GPT-3 in InstructGPT) trained to output scalar reward:

$$L(\theta) = -\frac{1}{N} \mathbb{E}_{(x,D,B) \sim D} [\log (\underbrace{\sigma(r_{\theta}(x, D))}_{\text{predicted reward for response D}} - \underbrace{r_{\theta}(x, B)}_{\text{predicted reward for response B}})]$$

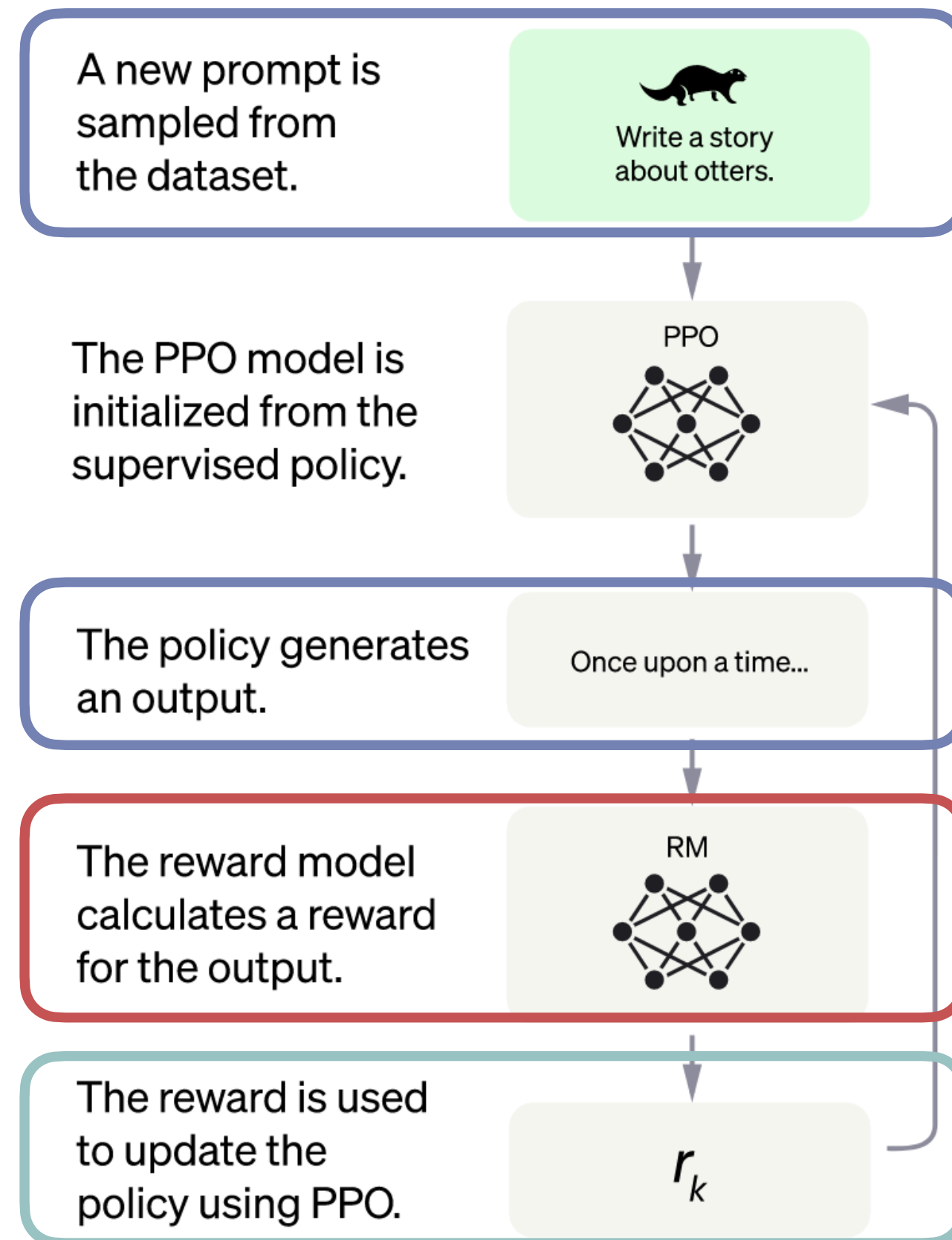
- ▶ smart procedure for eliciting comparisons

RLHF in practice

Step 3

Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.



- ▶ the model (= policy π) is adjusted to **maximize return**
- ▶ human **preferences encoded in the reward model** are used to provide the reward
 - RL training used to learn the policy maximizing the reward
 - maximizing the reward approximates receiving the **best feedback from humans**
- ▶ training via **Proximal Policy Optimization (PPO)** with bells & whistles

What just happened!?

- Supervised LM Finetuning
- Instruction Finetuning
- General RL
- RLHF

What is to come...

- H1 due today
- H2 released tomorrow
- Tutorial **May 15th**
- Next lecture May 20**th**
 - Agents
 - Assistants
 - Neuro-Symbolic Models